

Convex Aggregation Certificates for Quadratic Systems

1 Quadratic systems and the certificate question

Let $n, m \geq 1$, and let

$$f_i(x) = x^\top A_i x + 2b_i^\top x + c_i, \quad S = \{x \in \mathbb{R}^n : f_i(x) < 0 \text{ for } i = 1, \dots, m\}, \quad (1.1)$$

where $A_i \in \mathbb{S}^n$, $b_i \in \mathbb{R}^n$, and $c_i \in \mathbb{R}$. Here $\mathbb{S}^n$ denotes the real symmetric $n \times n$ matrices; $A \succeq 0$ and $A \succ 0$ mean positive semidefinite and positive definite, respectively. We use the Euclidean vector norm and the Frobenius matrix norm, and write $\langle A, B \rangle = \text{tr}(AB)$. The symbol $\text{conv } S$ denotes the ordinary convex hull, without taking its closure. A hull is *proper* if it differs from $\mathbb{R}^n$. Our question is whether a nonempty quadratic system has a proper hull, and how this can be certified by its defining inequalities.

For $\lambda \in \mathbb{R}^m$, write

$$(A_\lambda, b_\lambda, c_\lambda) = \sum_{i=1}^m \lambda_i (A_i, b_i, c_i), \quad f_\lambda = \sum_{i=1}^m \lambda_i f_i.$$

A nonzero $\lambda \geq 0$ defines an *aggregation*: $f_\lambda(x) < 0$ is valid for every $x \in S$. We call λ a *convex certificate* if $A_\lambda \succeq 0$, so that f_λ is convex on $\mathbb{R}^n$. It is *nontrivial* if $(A_\lambda, b_\lambda) \neq (0, 0)$. Otherwise the aggregated function is constant. If $S \neq \emptyset$, that constant is strictly negative, as evaluation at any point of S shows. We retain zero in the following cone for convenience:

$$\mathcal{K} = \{\lambda \in \mathbb{R}_+^m : A_\lambda \succeq 0\}, \quad \mathcal{T} = \{\lambda \in \mathbb{R}^m : A_\lambda = 0, b_\lambda = 0\}. \quad (1.2)$$

Thus a nontrivial convex certificate exists precisely when $\mathcal{K} \not\subseteq \mathcal{T}$.

The homogeneous matrices and map associated with (1.1) are

$$Q_i = \begin{pmatrix} A_i & b_i \\ b_i^\top & c_i \end{pmatrix}, \quad f_i^h(x, t) = x^\top A_i x + 2tb_i^\top x + c_i t^2, \quad f^h = (f_1^h, \dots, f_m^h). \quad (1.3)$$

We also write $Q_\lambda = \sum_i \lambda_i Q_i$ and $g_i(x) = x^\top A_i x$. Define the strict homogeneous feasible set $S^h = \{(x, t) : f_i^h(x, t) < 0 \text{ for all } i\}$. It is open and invariant under multiplication by every nonzero real scalar. For $t \neq 0$, $(x, t) \in S^h$ is equivalent to $x/t \in S$. For $\alpha \in \mathbb{R}^n \setminus \{0\}$ and $s \in \mathbb{R}$, put

$$H_{\alpha, s} = \{(x, t) \in \mathbb{R}^{n+1} : \alpha^\top x = st\}, \quad E = \{(x, t) : t = 0\}. \quad (1.4)$$

Definition 1.1 (Hidden hyperplane convexity). A homogeneous quadratic map $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ has *hidden hyperplane convexity* (HHC) if $\phi(H)$ is convex for every linear hyperplane $H \subset \mathbb{R}^d$.

This is the definition of Blekherman, Dey, and Sun [7, Definition 2.1]; their work appeared as [8]. Ordinary *hidden convexity* means only that $\phi(\mathbb{R}^d)$ is convex. The two properties differ. All homogeneous quadratic images contain zero and are invariant under nonnegative scaling, so a convex such image is a convex cone; it need not be closed. HHC implies ordinary hidden convexity when $d \geq 3$: any two points of $\mathbb{R}^d$ lie in some linear hyperplane, whose image contains the segment joining their images.

Question 1.2 (Blekherman–Dey–Sun, Conjecture 3.3). Suppose $S \neq \emptyset$ and f^h has HHC. Is

$$\text{conv } S \neq \mathbb{R}^n \iff \exists \lambda \in \mathbb{R}_+^m : A_\lambda \succeq 0, \quad (A_\lambda, b_\lambda) \neq (0, 0)? \quad (1.5)$$

The formulation in Blekherman et al. [7, Conjecture 3.3] says that the aggregated function is not a negative constant. Under nonemptiness this is equivalent to the displayed condition. Their standing setting has $n \geq 3$ and $m \geq 2$; the formulation here also includes the smaller positive dimensions and numbers of inequalities. Throughout, numbered result locators for that work refer to the explicitly cited May 29, 2023 arXiv version; the earlier author manuscript has different numbering for several background results.

The reverse implication in (1.5) is elementary: a nonconstant convex quadratic aggregation bounds S by a proper convex sublevel set. The unresolved case in the cited conjecture concerns the opposite implication when $\mathcal{K}$ contains nonzero vectors but all of them belong to $\mathcal{T}$. Under convexity of $(g_1, \dots, g_m)(\mathbb{R}^n)$, Blekherman, Dey, and Sun already prove that $\mathcal{K} = \{0\}$ implies $\text{conv } S = \mathbb{R}^n$ [7, Proposition 2.14]. For $n \geq 2$, their separable case, with all Q_i diagonal, also already gives the certificate characterization [7, Theorem 2.12].

It is essential to distinguish this certificate question from a complete description of the hull. In the aggregation literature, a *good aggregation* is a nonnegative nonzero multiplier λ such that $\text{conv } S \subseteq \{x : f_\lambda(x) < 0\}$ and Q_λ has at most one negative eigenvalue. The function f_λ need not be globally convex. Blekherman, Dey, and Sun prove that these good aggregations describe $\text{conv } S$ under HHC, $n \geq 3$, and $\emptyset \neq S$ with proper hull [7, Theorem 2.9]. Thus an existence certificate and a family of hull-defining aggregations address different questions.

The two-constraint case belongs to established theory. Dines’ theorem gives convexity of the joint image of two homogeneous quadratic forms [14], and therefore HHC for a pair of homogeneous forms and their restrictions. Yildiran characterizes the hull of a nonempty set defined by two strict quadratic inequalities using at most two aggregations [26]. For three quadratic inequalities in dimension $n \geq 3$ with a nonempty proper hull, Dey, Muñoz, and Serrano obtain an aggregation description when the homogenized matrices Q_1, Q_2, Q_3 admit a positive definite real linear combination [12]. Here “positive definite linear combination” allows real coefficients of either sign; it does not mean a nonnegative aggregation. More recently, Blekherman and Dunbar use the spectral curve and algebraic topology to obtain further three-quadratic hull descriptions [6]. Their available preprint treats finiteness and related spectral conditions [5]. These results set the context for the certificate question, while neither the classical two-form convexity theorem nor an elementary separation lemma is itself a new contribution of the present work.

2 The certificate theorem

Only a family of hyperplanes approaching E is needed to answer Question 1.2. We first make this weaker assumption precise.

Definition 2.1 (Asymptotic hyperplane convexity). The map f^h has *asymptotic hyperplane convexity* (AHC) if, for every $\alpha \neq 0$ and every $R \in \mathbb{R}$, there exists $s > R$ such that $f^h(H_{\alpha,s})$ is convex.

Equivalently, for each $\alpha \neq 0$ there is a strictly increasing sequence (s_k) tending to $+\infty$ with all these images convex. Indeed, the sequence condition implies the unbounded-level condition; conversely, choose a good $s_1 > 1$ and recursively a good $s_{k+1} > \max\{s_k, k+1\}$. HHC implies AHC because $H_{\alpha,s}$ is a linear hyperplane for every s . This assumption concerns the images themselves and requires no closedness.

Theorem 2.2 (Convex aggregation certificate). *Suppose $S \neq \emptyset$ and f^h has AHC. Then*

$$\text{conv } S \neq \mathbb{R}^n \iff \exists \lambda \geq 0 : \quad A_\lambda \succeq 0, \quad (A_\lambda, b_\lambda) \neq (0, 0). \quad (2.1)$$

The certificate-to-proper-hull implication holds without either assumption.

Corollary 2.3 (Conjecture 3.3 of Blekherman–Dey–Sun). *If $S \neq \emptyset$ and f^h has HHC, then (2.1) holds.*

The corollary follows from HHC implying AHC. It answers the cited conjecture for every finite number of inequalities, including the case where nonzero convex certificates exist but all their quadratic and linear coefficients might vanish. The proof below shows that this case forces a whole-space hull. The theorem also isolates a weaker sufficient hypothesis; we attach no novelty claim to the separation or compactness facts it uses.

2.1 Hyperplanes beyond a supporting halfspace

Lemma 2.4 (No common strict negative quadratic direction). *If $\text{conv } S \neq \mathbb{R}^n$, then there is no $v \in \mathbb{R}^n$ satisfying $v^\top A_i v < 0$ for every i . Equivalently, $E \cap S^h = \emptyset$.*

Proof. If such a v existed, then for any fixed $y \in \mathbb{R}^n$,

$$f_i(y + rv) = r^2 v^\top A_i v + 2r(v^\top A_i y + b_i^\top v) + f_i(y) \longrightarrow -\infty \quad (|r| \longrightarrow \infty).$$

There are finitely many constraints. Hence both $y + rv$ and $y - rv$ belong to S for all sufficiently large positive r , and their midpoint y belongs to $\text{conv } S$. This holds for every y , a contradiction. Finally, $f_i^h(v, 0) = v^\top A_i v$ proves the equivalence. $\square$

This lemma concerns a simultaneous strict sign condition on the leading forms. It is not a characterization of the recession cone of S .

Lemma 2.5 (Sweeping hyperplanes). *Suppose $S \neq \emptyset$ and $\text{conv } S \neq \mathbb{R}^n$. There are $\alpha \neq 0$ and $\beta \in \mathbb{R}$ such that*

$$\alpha^\top x < \beta \quad (x \in S). \quad (2.2)$$

For any such α, β , one has $H_{\alpha, s} \cap S^h = \emptyset$ for every $s \geq \beta$.

Proof. The set S is open, and its convex hull is open as well: in a finite convex combination, one point with positive weight can vary in an open ball, giving an open ball around the combination. Separate a point $y \notin \text{conv } S$ from this nonempty open convex set. Finite-dimensional separation supplies $\alpha \neq 0$ with $\alpha^\top x \leq \alpha^\top y$ throughout $\text{conv } S$. Equality at a point of that open set would be contradicted by a small move in direction α . Thus (2.2) holds with $\beta = \alpha^\top y$.

If $(x, t) \in H_{\alpha, s} \cap S^h$ and $t = 0$, then Lemma 2.4 gives a contradiction. If $t \neq 0$, the identity $f_i^h(x, t) = t^2 f_i(x/t)$ gives $x/t \in S$, whereas $\alpha^\top(x/t) = s \geq \beta$, again a contradiction. This reasoning applies to either sign of t . $\square$

Lemma 2.6 (Separation on a hyperplane). *Suppose $s \neq 0$, the set $f^h(H_{\alpha, s})$ is convex, and $H_{\alpha, s} \cap S^h = \emptyset$. There exists $\lambda \geq 0$ with $\sum_i \lambda_i = 1$ such that, for every $x \in \mathbb{R}^n$,*

$$0 \leq x^\top A_\lambda x + \frac{2}{s}(\alpha^\top x)(b_\lambda^\top x) + \frac{c_\lambda}{s^2}(\alpha^\top x)^2. \quad (2.3)$$

Proof. The convex image $C = f^h(H_{\alpha,s})$ contains zero and is disjoint from the open negative orthant $N = -\mathbb{R}_{++}^m$. Separation of these two convex sets gives a nonzero λ and a real a with $\lambda^\top z \leq a \leq \lambda^\top y$ for $z \in N$, $y \in C$. If some $\lambda_i < 0$, sending z_i to $-\infty$ with the other coordinates fixed contradicts the upper bound. Thus $\lambda \geq 0$, and $\sup_{z \in N} \lambda^\top z = 0$. Since $0 \in C$, necessarily $a = 0$, and $\lambda^\top f^h \geq 0$ on $H_{\alpha,s}$. Divide λ by its positive coordinate sum. Substituting $(x, t) = (x, \alpha^\top x/s)$ gives (2.3). Closedness of C was not used. $\square$

2.2 A nonzero limit in coefficient space

The relevant cone retains the quadratic and linear coefficients together:

$$P = \text{cone}\{(A_i, b_i) : 1 \leq i \leq m\} = \{(A_\lambda, b_\lambda) : \lambda \geq 0\} \subseteq \mathbb{S}^n \times \mathbb{R}^n. \quad (2.4)$$

We equip this space with $\|(A, b)\| = (\|A\|_F^2 + \|b\|^2)^{1/2}$. Closedness of this particular cone is essential: a limit in its closure must correspond to actual aggregation weights. The following elementary fact supplies that property.

Lemma 2.7 (Finite coefficient cones are closed). *The cone generated by a finite family in a finite-dimensional real vector space is closed.*

Proof. Every conic combination can be represented using a linearly independent subfamily. To see this, choose a representation with the fewest positive coefficients. If its active generators are dependent, take a nonzero relation $\sum_i d_i v_i = 0$, reversing its sign so some $d_i > 0$. For the active coefficients $a_i > 0$, set $\tau = \min_{d_i > 0} a_i/d_i$. Then all $a_i - \tau d_i$ are nonnegative, at least one vanishes, and the represented point is unchanged, a contradiction. For a linearly independent subfamily, the coefficient map is a linear homeomorphism onto its span; that span is closed, so the image of the nonnegative orthant is closed. There are only finitely many subfamilies. The original cone is their finite union and is closed. The empty subfamily accounts for the zero vector. $\square$

Proof of Theorem 2.2. First suppose a nontrivial convex certificate λ exists. Its nonconstant coefficient pair implies $\lambda \neq 0$, so $f_\lambda < 0$ on S . Positive semidefiniteness makes f_λ convex, whence $\text{conv } S \subseteq \{f_\lambda < 0\}$. This sublevel set is proper. Indeed, if $A_\lambda \neq 0$, it has a unit eigenvector v with eigenvalue $\mu > 0$, and $f_\lambda(rv) = \mu r^2 + 2rb_\lambda^\top v + c_\lambda \rightarrow +\infty$ as $r \rightarrow +\infty$. If $A_\lambda = 0$, then $b_\lambda \neq 0$ and $f_\lambda(rb_\lambda) = 2r\|b_\lambda\|^2 + c_\lambda \rightarrow +\infty$. This proves the easy implication without using nonemptiness or AHC.

For the converse, suppose $S \neq \emptyset$, f^h has AHC, and $\text{conv } S \neq \mathbb{R}^n$. Fix $x_0 \in S$, and choose α, β from Lemma 2.5. AHC gives positive levels $s_k \rightarrow +\infty$ with $s_k \geq \beta$ and convex images $f^h(H_{\alpha,s_k})$. Lemmas 2.5 and 2.6 give nonzero normalized weights $\lambda^k \geq 0$ satisfying (2.3). Write $(A_k, b_k, c_k) = (A_{\lambda^k}, b_{\lambda^k}, c_{\lambda^k})$. Strict feasibility gives the crucial upper bound

$$c_k < -x_0^\top A_k x_0 - 2b_k^\top x_0 =: L(A_k, b_k), \quad (2.5)$$

where L is a fixed continuous linear functional on coefficient space. The multiplier of c_k in (2.3) is a square. Replacing c_k by its upper bound therefore yields

$$0 \leq x^\top A_k x + \frac{2}{s_k}(\alpha^\top x)(b_k^\top x) + \frac{L(A_k, b_k)}{s_k^2}(\alpha^\top x)^2 \quad (x \in \mathbb{R}^n). \quad (2.6)$$

Moreover, $(A_k, b_k) \neq (0, 0)$: otherwise (2.5) gives $c_k < 0$, and the original inequality (2.3) at $x = \alpha$ reads $0 \leq c_k \|\alpha\|^4 / s_k^2 < 0$.

Set $\rho_k = \|(A_k, b_k)\| > 0$ and $(\hat{A}_k, \hat{b}_k) = (A_k, b_k)/\rho_k$. These pairs lie in the unit section of P , which is compact by Lemma 2.7. Passing to a subsequence, let

$$(\hat{A}_k, \hat{b}_k) \longrightarrow (A_*, b_*) \in P, \quad \|(A_*, b_*)\| = 1.$$

Divide (2.6) by ρ_k . By linearity of L , its right-hand side becomes

$$x^\top \hat{A}_k x + \frac{2}{s_k} (\alpha^\top x) (\hat{b}_k^\top x) + \frac{L(\hat{A}_k, \hat{b}_k)}{s_k^2} (\alpha^\top x)^2.$$

For every fixed x the last two terms tend to zero. Consequently $x^\top A_* x \geq 0$ for every x . This conclusion uses one coefficient subsequence for all x ; no point-dependent subsequence is chosen. The symmetric matrix A_* is positive semidefinite. Since $(A_*, b_*) \in P$, there are actual weights $\lambda^* \geq 0$ with $(A_{\lambda^*}, b_{\lambda^*}) = (A_*, b_*)$. Their coefficient pair has norm one, so they give the required nontrivial convex certificate. $\square$

Remark 2.8 (What the limit proves). Normalizing the weights in the simplex alone can give a negative constant aggregation in the limit. Normalizing the nonconstant coefficient pair prevents this loss of information. Bound (2.5) removes the constant before normalization; no bound on c_k/ρ_k is assumed. The final weights need not be a limit of the original weights, or of their rescalings. Their existence follows from closedness of the finitely generated cone P .

Remark 2.9 (Convexity along one normal). The hard implication needs convexity of $f^h(H_{\alpha,s})$ at arbitrarily large levels for only one α satisfying (2.2). AHC is a convenient condition that can be stated without knowing a supporting normal. For fixed α , the hyperplanes $H_{\alpha,s}$ approach E as $s \rightarrow +\infty$; convexity of $f^h(E)$ alone is not the hypothesis used in the proof. There is no restriction beyond $n, m \geq 1$. For one constraint, a negative quadratic direction gives a whole-space hull by Lemma 2.4; the remaining cases follow from the easy implication or a negative constant. For two constraints, Dines' theorem makes HHC automatic, recovering the known certificate case discussed in Section 1.

2.3 A quantitative form of the obstruction

For completeness, we give a second way to exclude degeneration to constant certificates. It supplies a uniform bound on the sweep level under the hypothesis that all convex certificates are trivial. The cone estimate itself is a standard compactness argument.

Lemma 2.10 (Uniform cone separation). *Let P, D be closed cones containing zero in a finite-dimensional normed space, with $P \cap D = \{0\}$. Then some $\kappa > 0$ satisfies*

$$\text{dist}(p, D) \geq \kappa \|p\| \quad (p \in P).$$

Neither cone is required to be convex.

Proof. If $P = \{0\}$, any positive κ works. Otherwise the continuous function $p \mapsto \text{dist}(p, D)$ has a positive minimum κ on the compact set $P \cap \{\|p\| = 1\}$: a zero distance would put a unit point of P in the closed set D . Scaling both cones proves the bound for $p \neq 0$, and the assertion at zero follows from $0 \in D$. $\square$

Apply the lemma to P in (2.4) and $D = \mathbb{S}_+^n \times \mathbb{R}^n$. The condition $P \cap D = \{0\}$ is exactly the assertion that every convex certificate is trivial. If A_- is the positive semidefinite negative part of a symmetric matrix A , then

$$\text{dist}((A, b), D) = \|A_-\|_F \leq \sqrt{n} \max\{0, -\lambda_{\min}(A)\}. \quad (2.7)$$

For the equality, diagonalize A orthogonally. In that basis every positive semidefinite approximant has nonnegative diagonal entries; the sum of squared errors is at least the sum of squares of the negative eigenvalues of A . Replacing those eigenvalues by zero achieves this bound, and the b component can be left unchanged. The inequality follows by bounding each negative eigenvalue in absolute value by $\max\{0, -\lambda_{\min}(A)\}$.

Proposition 2.11 (Bound on a sweep certificate). *Suppose every convex certificate is trivial, and let κ be as in Lemma 2.10 for the preceding cones. If $\lambda \geq 0$, $c_\lambda < 0$, $s > 0$, $\alpha \neq 0$, and (2.3) holds, then*

$$s \leq \frac{2\sqrt{n}\|\alpha\|}{\kappa}. \quad (2.8)$$

Proof. The inequality at $x = \alpha$ shows that $\rho = \|(A_\lambda, b_\lambda)\|$ cannot vanish. The cone estimate and (2.7) give $\lambda_{\min}(A_\lambda) \leq -\kappa\rho/\sqrt{n}$. Choose a unit eigenvector u for this eigenvalue. Evaluating (2.3) at u , discarding the nonpositive constant term, and using $\|b_\lambda\| \leq \rho$ gives

$$0 \leq -\frac{\kappa}{\sqrt{n}}\rho + \frac{2\|\alpha\|}{s}\rho.$$

Division by $\rho > 0$ proves (2.8). $\square$

This gives an alternative completion of the hard implication. Under its hypotheses, suppose every convex certificate is trivial and choose the normalized sweep weights λ^k as before. Compactness of the simplex gives a subsequence converging to $\bar{\lambda}$, with $\bar{\lambda} \geq 0$ and $\sum_i \bar{\lambda}_i = 1$. The coefficients b_k, c_k are bounded, so letting $k \rightarrow \infty$ in (2.3) shows $A_{\bar{\lambda}} \succeq 0$. Triviality then gives $A_{\bar{\lambda}} = 0$, $b_{\bar{\lambda}} = 0$, and $c_{\bar{\lambda}} < 0$ by strict feasibility. Thus $c_k < 0$ eventually along this subsequence. Proposition 2.11 contradicts $s_k \rightarrow +\infty$. The constant κ depends only on the original quadratic and linear coefficients; this argument neither bounds aggregation weights nor provides a procedure for verifying AHC.

3 Consequences for closed systems and semidefinite relaxations

The certificate theorem determines whether a hull is the whole space. Its consequences below concern that dichotomy; they do not identify every supporting inequality of a proper hull.

3.1 Closed inequalities and the three feasibility regimes

Write $T = \{x : f_i(x) \leq 0 \text{ for all } i\}$.

Corollary 3.1. *Suppose $S \neq \emptyset$ and f^h has asymptotic hyperplane convexity. Then*

$$\text{conv } T \neq \mathbb{R}^n \iff \text{conv } S \neq \mathbb{R}^n \iff \mathcal{K} \not\subseteq \mathcal{T}.$$

Proof. The inclusion $S \subseteq T$ gives one implication. A nontrivial convex certificate bounds T by the closed convex set $\{f_\lambda \leq 0\}$. That set is proper because the nonconstant convex quadratic f_λ is unbounded above. The remaining implication is Theorem 2.2. $\square$

Strict feasibility is essential here; Example 4.5 gives a counterexample even with HHC and a closed-system hull with interior. No equality between $\text{conv } T$ and $\text{cl conv } S$ is asserted.

Corollary 3.2 (Three regimes under HHC). *Suppose f^h has HHC and $n \geq 2$. Exactly one of the following holds:*

1. $S = \emptyset$. This is equivalent to the existence of $\lambda \geq 0$, $\lambda \neq 0$, with $Q_\lambda \succeq 0$.
2. $S \neq \emptyset$ and $\text{conv } S = \mathbb{R}^n$. This is equivalent to nonemptiness together with $\mathcal{K} \subseteq \mathcal{T}$.
3. $S \neq \emptyset$ and $\text{conv } S \neq \mathbb{R}^n$. This is equivalent to nonemptiness together with $\mathcal{K} \not\subseteq \mathcal{T}$. If also $n \geq 3$, the good aggregations defined in Section 1 describe $\text{conv } S$ exactly.

Proof. For the first statement, HHC implies convexity of $f^h(\mathbb{R}^{n+1})$ because $n+1 \geq 3$. Moreover, $S = \emptyset$ if and only if $S^h = \emptyset$: a strict homogeneous point with $t = 0$ can be perturbed to one with $t \neq 0$, and then dehomogenized. Separation of the convex homogeneous image from the open negative orthant gives a nonzero $\lambda \geq 0$ for which $\lambda^\top f^h \geq 0$, equivalently $Q_\lambda \succeq 0$. The converse follows by evaluation at a hypothetical strict point. This is the classical separation argument used in Blekherman et al. [7, Proposition 2.13]. Theorem 2.2 supplies the second and third statements. The final hull description is Blekherman et al. [7, Theorem 2.9]. $\square$

The proper-hull certificate itself remains valid for $n = 1$ by Theorem 2.2; the dimension restrictions above belong to the additional emptiness and full-description statements.

3.2 A finite semidefinite characterization

The certificate cone can be examined on the compact slice

$$\mathcal{D} = \{\lambda \geq 0 : \sum_i \lambda_i = 1, A_\lambda \succeq 0\}.$$

If $\mathcal{D}$ is nonempty, let

$$\tau = \max_{\lambda \in \mathcal{D}} \text{tr } A_\lambda, \quad \beta_j^\pm = \max_{\lambda \in \mathcal{D}} (\pm (b_\lambda)_j) \quad (j = 1, \dots, n). \quad (3.1)$$

All maxima are attained. These are $2n+1$ semidefinite programs, with one $n \times n$ positive semidefinite block and m nonnegative weights.

Corollary 3.3. *The inclusion $\mathcal{K} \subseteq \mathcal{T}$ holds exactly when $\mathcal{D} = \emptyset$, or $\mathcal{D} \neq \emptyset$ and all the values in (3.1) are zero. Under the hypotheses of Theorem 2.2, these conditions characterize $\text{conv } S = \mathbb{R}^n$.*

Proof. Every nonzero element of $\mathcal{K}$ has a positive multiple in $\mathcal{D}$. For a positive semidefinite matrix, zero trace is equivalent to the matrix being zero. Thus $\tau = 0$ forces $A_\lambda = 0$ throughout $\mathcal{D}$. The two coordinate objectives force $(b_\lambda)_j \leq 0$ and $(b_\lambda)_j \geq 0$ there. The converse is immediate. If $\mathcal{D}$ is empty, $\mathcal{K} = \{0\}$. $\square$

The reduction uses trace in place of all matrix coordinates. It is an exact characterization by feasibility and optimal values, not a bit-complexity result or a floating-point decision procedure. In particular, small computed objective values do not establish the exact zeros needed for a whole-space conclusion. Feasibility of $\mathcal{D}$ can be checked as the common feasibility part of these programs.

3.3 What the Shor projection preserves

Define the projection of the standard semidefinite relaxation by

$$P_{\text{SDP}} = \left\{ x \in \mathbb{R}^n : \begin{array}{l} \exists X \in \mathbb{S}^n, \quad X - xx^\top \succeq 0, \\ \langle A_i, X \rangle + 2b_i^\top x + c_i \leq 0 \quad (i = 1, \dots, m) \end{array} \right\}. \quad (3.2)$$

The matrix condition is equivalent to $\begin{pmatrix} 1 & x^\top \\ x & X \end{pmatrix} \succeq 0$, so P_{SDP} is convex. It contains T , by taking $X = xx^\top$. Let

$$C_{\text{agg}} = \bigcap_{\lambda \in \mathcal{K}} \{x : f_\lambda(x) \leq 0\}. \quad (3.3)$$

This is a closed convex set. For any Shor witness X and $\lambda \in \mathcal{K}$,

$$f_\lambda(x) = \sum_i \lambda_i (\langle A_i, X \rangle + 2b_i^\top x + c_i) - \langle A_\lambda, X - xx^\top \rangle \leq 0. \quad (3.4)$$

Consequently $P_{\text{SDP}} \subseteq C_{\text{agg}}$. This is the classical semidefinite/convex-aggregation relationship studied by Fujie and Kojima [17], Kojima and Tunçel [18]. We prove the precise closure statement needed here; projection closedness must not be assumed.

Proposition 3.4. *If $S \neq \emptyset$, then $\text{cl } P_{\text{SDP}} = C_{\text{agg}}$. Under the same assumption, without any hidden-convexity hypothesis,*

$$P_{\text{SDP}} = \mathbb{R}^n \iff \mathcal{K} \subseteq \mathcal{T}. \quad (3.5)$$

Proof. Put $g(v) = (v^\top A_i v)_i$ and introduce the convex cone

$$\mathcal{C} = \{(\langle A_i, Y \rangle)_i : Y \succeq 0\} + \mathbb{R}_+^m.$$

It contains $\mathbb{R}_+^m$ and is therefore full dimensional. It need not be closed. Directly from (3.2),

$$x \in P_{\text{SDP}} \iff -f(x) \in \mathcal{C}, \quad \mathcal{C}^* = \mathcal{K}, \quad x \in C_{\text{agg}} \iff -f(x) \in \text{cl } \mathcal{C}.$$

The dual identity follows from the self-duality of the positive semidefinite cone; the last equivalence is the finite-dimensional bipolar theorem.

We will use $\text{cl } \mathcal{C} + \text{int } \mathcal{C} \subseteq \text{int } \mathcal{C}$. For completeness, if $B(c, r) \subseteq \mathcal{C}$ and $d_k \in \mathcal{C}$ tends to $d \in \text{cl } \mathcal{C}$, then for large k the ball $B(c + d, r/2)$ lies in $B(c + d_k, r)$, which lies in $\mathcal{C} + \mathcal{C} = \mathcal{C}$. This also proves that adding another element of $\mathcal{C}$ preserves the interior.

Choose $x_0 \in S$, so $-f(x_0) \in \mathbb{R}_{++}^m \subseteq \text{int } \mathcal{C}$. For $x \in C_{\text{agg}}$ and $0 < t < 1$, put $x_t = (1 - t)x + tx_0$. The quadratic identity

$$-f(x_t) = (1 - t)(-f(x)) + t(-f(x_0)) + t(1 - t)g(x - x_0) \quad (3.6)$$

puts $-f(x_t)$ in $\text{int } \mathcal{C}$, since $g(v) = (\langle A_i, vv^\top \rangle)_i \in \mathcal{C}$. Thus $x_t \in P_{\text{SDP}}$, and $x_t \rightarrow x$ proves $C_{\text{agg}} \subseteq \text{cl } P_{\text{SDP}}$. The opposite inclusion follows from (3.4) and closedness of C_{agg} .

A nontrivial convex certificate makes C_{agg} proper, so $P_{\text{SDP}} = \mathbb{R}^n$ implies $\mathcal{K} \subseteq \mathcal{T}$. Conversely, suppose every certificate is trivial. For any $\lambda \in \mathcal{K} \setminus \{0\}$ strict feasibility gives $f_\lambda \equiv c_\lambda < 0$. Hence $C_{\text{agg}} = \mathbb{R}^n$ and $-f(z) \in \text{cl } \mathcal{C}$ for every z . For an arbitrary target x , apply (3.6) with $z = 2x - x_0$ in place of x and $t = 1/2$. Its left side is $-f(x)$; its right side belongs to $\text{int } \mathcal{C}$. Thus $x \in P_{\text{SDP}}$. $\square$

The closure conclusion also follows from Fujie and Kojima [17, Theorem 2.1]: strict feasibility provides their semidefinite Slater condition by choosing $X = x_0 x_0^\top + \varepsilon I$ for sufficiently small $\varepsilon > 0$. The elementary cone proof above records explicitly where strict feasibility enters and why no closedness of a cone image is required.

Corollary 3.5. *If $S \neq \emptyset$ and f^h has asymptotic hyperplane convexity, then*

$$P_{\text{SDP}} = \mathbb{R}^n \iff \text{conv } S = \mathbb{R}^n \iff \text{conv } T = \mathbb{R}^n.$$

Proof. Combine Proposition 3.4, Theorem 2.2, and Corollary 3.1. $\square$

Thus the relaxation preserves the existence of at least one nonconstant valid affine bound under these hypotheses. It need not preserve each particular bound; Example 4.6 makes the distinction explicit.

Example 3.6 (A nonclosed Shor projection with strict feasibility). In $\mathbb{R}^2$, let

$$f_1(x) = 2x_1x_2 + 1, \quad f_2(x) = x_1^2 - x_1.$$

The point $(1/2, -2)$ is strictly feasible. A nonnegative aggregate has quadratic matrix $\begin{pmatrix} \lambda_2 & \lambda_1 \\ \lambda_1 & 0 \end{pmatrix}$, which is positive semidefinite exactly when $\lambda_1 = 0$. Therefore

$$C_{\text{agg}} = [0, 1] \times \mathbb{R}, \quad P_{\text{SDP}} = ((0, 1) \times \mathbb{R}) \cup (\{1\} \times (-\infty, -1/2]).$$

To verify the latter equality, write $Y = X - xx^\top \succeq 0$. The second lifted row gives $0 \leq Y_{11} \leq x_1 - x_1^2$, so $0 \leq x_1 \leq 1$. At either endpoint $Y_{11} = 0$, and positive semidefiniteness forces $Y_{12} = 0$. At $x_1 = 0$ the first row then fails; at $x_1 = 1$ it requires $x_2 \leq -1/2$, and $Y = 0$ suffices. For $0 < x_1 < 1$, take

$$X_{11} = \frac{x_1^2 + x_1}{2}, \quad X_{12} = -\frac{1}{2}, \quad Y_{22} = \frac{(-1/2 - x_1x_2)^2}{(x_1 - x_1^2)/2}, \quad X_{22} = x_2^2 + Y_{22}.$$

Then $Y \succeq 0$ has zero determinant and the lifted rows hold. The displayed description proves nonclosedness even with strict feasibility and, by Dines' theorem, HHC.

Compactness of the original closed feasible set does not in general restore projection closedness. Add the two rows $f_3 = -2x_1x_2 - 2$ and $f_4 = 1/4 - x_1$ to this example. The point $(1/2, -3/2)$ is strictly feasible and

$$T = \{x : 1/4 \leq x_1 \leq 1, -1 \leq x_1x_2 \leq -1/2\}$$

is closed and bounded. A nonnegative aggregate is convex exactly when $\lambda_1 = \lambda_3$, and then equals

$$\lambda_2(x_1^2 - x_1) + \lambda_4(1/4 - x_1) - \lambda_1.$$

It follows that $C_{\text{agg}} = [1/4, 1] \times \mathbb{R}$. In contrast,

$$P_{\text{SDP}} = ([1/4, 1) \times \mathbb{R}) \cup (\{1\} \times [-1, -1/2]).$$

Indeed the linear fourth row requires $x_1 \geq 1/4$; for $x_1 < 1$ the preceding lift works with $X_{12} = -3/4$ and $Y_{22} = (-3/4 - x_1x_2)^2 / ((x_1 - x_1^2)/2)$. At $x_1 = 1$, $Y_{11} = Y_{12} = 0$, so the first and third rows require $-1 \leq x_2 \leq -1/2$, and a rank-one lift suffices. This compact variant is used only to demonstrate that a closure qualification is necessary; no HHC claim is needed for it.

In particular, our use of the classical SDP/aggregation relationship is the closure equality in Proposition 3.4 and Fujie and Kojima [17, Theorem 2.1]. We do not use the unqualified projection equality stated in Kojima and Tunçel [18, Theorem 4.2]; Example 3.6 explains the distinction, including when the original feasible set is compact. For that compact variant, the cone generated by its four defining quadratic functions is a closed convex cone by finite generation (Lemma 2.7). Replacing the finite list by this cone leaves the original feasible set, the semidefinite projection, and the convex-aggregation intersection unchanged: the original rows belong to the cone, and every cone member is a nonnegative combination of them. Thus it also meets the closed-convex-cone representation hypothesis in that theorem.

4 Checkable hypotheses and their limits

4.1 Stable convexity of the quadratic parts

A tuple $(A_1, \dots, A_m)$ is *stably convex* if it has a neighborhood in $(\mathbb{S}^n)^m$ such that every tuple $(M_1, \dots, M_m)$ in that neighborhood has a convex joint image $\{(x^\top M_i x)_i : x \in \mathbb{R}^n\}$. This is stable convexity in the sense of Sheriff [22], rather than convexity of each component function.

Proposition 4.1. *If the quadratic-part tuple $(A_i)_i$ is stably convex, then f^h has asymptotic hyperplane convexity, for arbitrary b_i, c_i . Consequently the certificate equivalence holds whenever $S \neq \emptyset$ and either*

1. $m = 2$, in any dimension $n \geq 1$; or
2. $m = 3$, $n \geq 3$, and some real combination $\sum_i \theta_i A_i$ is positive definite.

Proof. For $s \neq 0$, parameterizing $H_{\alpha, s}$ by $(x, \alpha^\top x/s)$ makes its image the joint image of

$$M_i(s) = A_i + s^{-1}(\alpha b_i^\top + b_i \alpha^\top) + s^{-2} c_i \alpha \alpha^\top. \quad (4.1)$$

For each fixed α , these matrices tend to A_i , so stable convexity gives convexity for all sufficiently large s . Dines' theorem gives convexity for every two-form tuple, including every perturbation and hyperplane restriction. For three forms on $\mathbb{R}^n$, $n \geq 3$, Polyak's theorem states that a positive definite real combination implies convexity of the joint image [19, Theorem 2.1]. Positive definiteness persists under small perturbations, so this gives stable convexity and the claimed conclusion. $\square$

The first case is the known two-constraint certificate setting. The second is a convenient criterion on the quadratic blocks, not a new three-form convexity theorem. One must also distinguish positive definiteness in $\text{span}\{A_i\}$ from positive definiteness in $\text{span}\{Q_i\}$. The latter implies the former, but in the delicate case of a trivial certificate they are equivalent. Indeed, if $\lambda^0 \geq 0$, $\lambda^0 \neq 0$, and $A_{\lambda^0} = b_{\lambda^0} = 0$, strict feasibility gives $c_{\lambda^0} < 0$. Thus $e_{n+1} e_{n+1}^\top = Q_{\lambda^0} / c_{\lambda^0}$ belongs to the real span of the Q_i . If $\sum_i \theta_i Q_i = \begin{pmatrix} A & b \\ b^\top & c \end{pmatrix}$ has $A \succ 0$, adding $r e_{n+1} e_{n+1}^\top$ with $r > b^\top A^{-1} b - c$ makes it positive definite by the Schur complement. If no nonzero trivial certificate exists, either a nontrivial one already exists or $\mathcal{K} = \{0\}$; under convexity of the quadratic-part image, the latter case was covered in Blekherman et al. [7, Proposition 2.14]. Thus the block formulation should not be read as resolving an additional previously open PDLC case beyond the main certificate theorem.

Example 4.2 (Asymptotic hyperplane convexity is strictly weaker than HHC). In $\mathbb{R}^3$, set

$$\begin{aligned} f_1 &= x_1^2 + x_2^2 + x_3^2 - 1, \\ f_2 &= x_1^2 - x_2^2 + x_3^2 - 1, \\ f_3 &= 2x_1 x_2 + x_3^2 - 1. \end{aligned}$$

The origin is strictly feasible. Since $A_1 = I$, the tuple is stably convex by the three-form result, and Proposition 4.1 gives asymptotic hyperplane convexity. On the homogeneous hyperplane $x_3 = t$, however, the image is

$$\{(x_1^2 + x_2^2, x_1^2 - x_2^2, 2x_1 x_2) : x_1, x_2 \in \mathbb{R}\}.$$

Every image point (y_1, y_2, y_3) satisfies $y_1^2 = y_2^2 + y_3^2$. The image contains $(1, 1, 0)$ and $(1, -1, 0)$ but not their midpoint $(1, 0, 0)$. Thus HHC fails. This also proves that stable convexity of the quadratic parts does not imply HHC of the homogenization.

Example 4.3 (HHC need not give stable convexity of the parts). Let $f_1 = f_2 = f_3 = x_1^2 - x_2^2 - 1$ in $\mathbb{R}^3$. On every homogeneous hyperplane the three components agree. A single real homogeneous quadratic form has image $\{0\}$, a ray, or $\mathbb{R}$, as follows by scaling and, in the indefinite case, evaluating both signs. Hence every such hyperplane image is convex. For any $\varepsilon > 0$, perturb the three quadratic parts to

$$u = x_1^2 - x_2^2, \quad u + \varepsilon(x_1^2 + x_2^2), \quad u + 2\varepsilon x_1 x_2.$$

Their values at e_1, e_2 have midpoint $(0, \varepsilon, 0)$. A preimage would require $x_1^2 = x_2^2 = 1/2$ and $x_1 x_2 = 0$, which is impossible. These arbitrarily small perturbations are nonconvex. Together with Example 4.2, this separates the two sufficient conditions in both directions.

4.2 Why ordinary hidden convexity is insufficient

Example 4.4. Consider, in $\mathbb{R}^3$ with x_3 free,

$$f_1 = x_1^2 - x_2^2 - 1, \quad f_2 = x_2^2 - x_1^2 - 1, \quad f_3 = x_1 x_2 - 1, \quad f_4 = -x_1 x_2 - 1. \quad (4.2)$$

The strict set contains the origin. On S ,

$$(x_1^2 + x_2^2)^2 = (x_1^2 - x_2^2)^2 + 4x_1^2 x_2^2 < 5.$$

It lies in the convex cylinder $\{x : x_1^2 + x_2^2 < \sqrt{5}\}$, so its hull is proper. Every aggregated quadratic matrix is traceless and every linear part is zero. Positive semidefiniteness therefore forces the matrix to vanish, so all convex certificates are trivial. More explicitly, the only conditions are $\lambda_1 = \lambda_2$ and $\lambda_3 = \lambda_4$.

Nevertheless both the full homogeneous image and the image on E are convex. Put $u = x_1^2 - x_2^2$, $v = x_1 x_2$, and $w = t^2$. The complex square map shows that (u, v) ranges over $\mathbb{R}^2$; w ranges independently over $\mathbb{R}_+$. Hence the full image is the linear image of $\mathbb{R}^2 \times \mathbb{R}_+$ under

$$(u, v, w) \mapsto (u - w, -u - w, v - w, -v - w).$$

The image on E is its restriction to $w = 0$ and is convex too.

For $s \neq 0$ restrict instead to $x_1 = st$. The same linear map is injective, and $w = x_1^2/s^2$. The two preimages $(x_1, x_2, x_3, t) = (1, 0, 0, 1/s)$ and $(0, 1, 0, 0)$ give a midpoint whose (u, v, w) coordinates would have to be $(0, 0, 1/(2s^2))$. They would force $x_1^2 = x_2^2 = 1/2$ and $x_1 x_2 = 0$, a contradiction. Thus no nonzero level in this family has a convex image. This is the precise obstruction to asymptotic hyperplane convexity. Proposition 3.4 also gives $P_{\text{SDP}} = \mathbb{R}^3$ here, although the true hull is bounded in two coordinates.

Three inequalities already suffice for a proper-hull obstruction: delete f_4 . The trace argument and both ordinary hidden-convexity statements remain valid. With $p = x_1 + x_2$ and $q = x_1 - x_2$, strict feasibility requires $|pq| < 1$ and $p^2 - q^2 < 4$. If $|p| \geq 3$, then $|q| < 1/3$ and $p^2 < 4 + 1/9$, a contradiction. Thus $|p| < 3$ is a proper affine bound. The set is unbounded, since $x_1 = -x_2$ is feasible for every value of x_1 . The three-coordinate map $(u, v, w) \mapsto (u - w, -u - w, v - w)$ is also injective, so the preceding hyperplane obstruction persists.

4.3 The role of strict feasibility for closed systems

Example 4.5. In $\mathbb{R}^3$, let

$$f_1 = x_1 x_2 - 1, \quad f_2 = -x_1 x_2 + 1, \quad f_3 = x_1^2 - x_2^2.$$

Then $S = \emptyset$, whereas

$$T = \{x : x_1 x_2 = 1, |x_1| \leq |x_2|\}$$

is nonempty and has $|x_1| \leq 1$. Consequently its hull is proper. The convex hull of the four feasible planar points $\pm(1, 1), \pm(1/2, 2)$ contains a neighborhood of the origin: the two positive points are linearly independent and their symmetric convex hull is a nondegenerate parallelogram. Since x_3 is free, $0 \in G := \text{int conv } T$ and G has full dimension.

The quadratic matrices are traceless and the linear parts vanish, so every convex certificate is trivial. HHC holds because $Q_2 = -Q_1$; every hyperplane image is a linear image of a two-form image, which is convex by Dines' theorem. This verifies all claimed properties without a sampled eigenvalue calculation.

This example also shows that the condition $Q_\lambda \neq 0$ for all nonzero $\lambda \geq 0$ in Blekherman et al. [7, Theorem 2.24 and Remark 2.25] cannot simply be removed. In their notation a good closed-system multiplier must satisfy $G \subseteq \{f_\lambda < 0\}$ and give Q_λ at most one negative eigenvalue. Since $0 \in G$, any such multiplier would have $c_\lambda = \lambda_2 - \lambda_1 < 0$. Put $\mu = \lambda_1 - \lambda_2 > 0$. Its homogeneous matrix is

$$Q_\lambda = \text{diag} \left(\begin{pmatrix} \lambda_3 & \mu/2 \\ \mu/2 & -\lambda_3 \end{pmatrix}, 0, -\mu \right).$$

The 2×2 block has determinant $-\lambda_3^2 - \mu^2/4 < 0$, so it contributes one negative eigenvalue; $-\mu$ contributes a second. There are therefore no good closed-system multipliers, whereas $\emptyset \neq G \neq \mathbb{R}^3$. Also $Q_{(1,1,0)} = 0$. Thus even taking an interior of the intersection of the good aggregations would give the whole space, not G .

4.4 A certificate need not describe the hull

Example 4.6. In $\mathbb{R}^3$ with x_2, x_3 free, let

$$f_1 = x_1^2 - 4, \quad f_2 = -x_1^2 + 2x_1 + 3.$$

Dines' theorem gives HHC. Factoring the two inequalities gives $S = \{x : -2 < x_1 < -1\}$, which is already convex. Globally convex nonzero aggregations have $\lambda_1 \geq \lambda_2 \geq 0$, with $\lambda_1 > 0$, and hence

$$f_\lambda(0) = -4\lambda_1 + 3\lambda_2 < 0.$$

Thus even all strict globally convex aggregations admit the origin, although $x_1 < -1$ is valid for S . The Shor projection also admits $x = 0$, with $X = \text{diag}(7/2, 0, 0)$: the two lifted residuals both equal $-1/2$. The relaxation is still proper, since the first convex quadratic inequality confines $|x_1| \leq 2$. Therefore the whole-space equivalence does not give hull exactness.

These examples distinguish the hypotheses, the open and closed systems, and the strength of the resulting certificate. The sufficient conditions in Proposition 4.1 supply concrete classes; no general procedure for verifying HHC or asymptotic hyperplane convexity is furnished by the certificate theorem.

5 A sharp Gram-map criterion for HHC

The certificate theorem separates existence of a useful aggregation from an exact description of the hull. To investigate the latter question, we first give a structural HHC criterion. It applies to any number of forms with repeated quadratic blocks and no linear terms. The matrix variational

identity used in its proof is classical; the conclusion concerns all real hyperplane restrictions of the full Gram map.

For positive integers k, r , consider

$$\Phi_{k,r} : \mathbb{R}^{k \times r} \times \mathbb{R} \longrightarrow \mathbb{S}^k \times \mathbb{R}, \quad \Phi_{k,r}(X, t) = (XX^\top, t^2).$$

We use the Frobenius inner product $\langle B, X \rangle = \text{tr}(B^\top X)$ for rectangular matrices. For $H, G \succeq 0$, define

$$F_H(G) = \left[\text{tr} \left((G^{1/2} H G^{1/2})^{1/2} \right) \right]^2. \quad (5.1)$$

This is the squared fidelity, also called transition probability, of the two positive semidefinite matrices, without a trace normalization.

Theorem 5.1 (Sharp Gram-map threshold). *The map $\Phi_{k,r}$ has HHC if and only if $r \geq k$. If $r \geq k$ and $(k, r) \neq (1, 1)$, then for every hyperplane*

$$L = \{(X, t) : \langle B, X \rangle + \beta t = 0\}, \quad (B, \beta) \neq (0, 0),$$

its image is exactly

$$\Phi_{k,r}(L) = \{(G, T) : G \succeq 0, T \geq 0, \beta^2 T \leq F_{BB^\top}(G)\}. \quad (5.2)$$

The formula includes singular G or $BB^\top$ and zero values of either hyperplane coefficient.

We prove the attainment and concavity statements separately. The following real trace-range argument explicitly handles the disconnectedness of the orthogonal group. Singular-value optimization over rotations is classical; see, for example, Ramachandran et al. [20, Lemma 1], which credits the determinant-corrected $SO(k)$ extremum to earlier work. Here an elementary rotation construction suffices.

Lemma 5.2 (Linear functionals on a Gram fiber). *Let $r \geq k$, $G \succeq 0$, and $B \in \mathbb{R}^{k \times r}$. The maximum of $\langle B, X \rangle$ over $XX^\top = G$ equals*

$$M = \text{tr} \left((G^{1/2} B B^\top G^{1/2})^{1/2} \right).$$

If $(k, r) \neq (1, 1)$, the attained values form the entire interval $[-M, M]$.

Proof. Every factor of G can be written as $X = G^{1/2} Q$ with $Q Q^\top = I_k$. Indeed, in an eigenbasis of G , normalize the rows belonging to positive eigenvalues and extend those orthonormal rows to k orthonormal rows in $\mathbb{R}^r$. Rows belonging to zero eigenvalues of G were already zero, so this extension preserves X .

Take a rectangular singular value decomposition $G^{1/2} B = U[D \ 0] V^\top$, where $D = \text{diag}(s_1, \dots, s_k)$ and $s_1 \geq \dots \geq s_k \geq 0$. Then

$$\langle B, X \rangle = \sum_{j=1}^k s_j (U^\top Q V)_{jj} \leq \sum_{j=1}^k s_j = M,$$

because every displayed diagonal entry has absolute value at most one. The choice $Q = U[I_k \ 0] V^\top$ attains M ; its negative attains $-M$.

For $k \geq 2$, restrict to $Q = U[R \ 0] V^\top$, with R orthogonal. If k is even, rotate each disjoint pair of coordinate axes through an angle $\theta \in [0, \pi]$. This gives a continuous path from I_k to $-I_k$, along

which $\text{tr}(DR)$ runs continuously from M to $-M$. If k is odd, then $k \geq 3$. Leave the last axis fixed and rotate the other pairs. The endpoint value is $2s_k - M \leq 0$, since $M \geq ks_k$. The intermediate value theorem therefore gives every value in $[0, M]$. Negating the factors gives $[-M, 0]$. If $k = 1$ and $r \geq 2$, the nonzero Gram fiber is a sphere in $\mathbb{R}^r$; a planar rotation from a maximizing vector to its negative gives the same conclusion. The zero fiber causes no exception. $\square$

Lemma 5.3 (Classical squared-fidelity formula). *For $G, H \succeq 0$,*

$$F_H(G) = \inf_{Z \succ 0} \text{tr}(GZ) \text{tr}(HZ^{-1}). \quad (5.3)$$

Consequently $G \mapsto F_H(G)$ is concave on $\mathbb{S}_+^k$ for fixed H .

Formula (5.3) and separate concavity are prior results in fidelity theory: see Uhlmann [23, pp. 408–410, equation (14) with the partial-fidelity index equal to zero]. We include a real-matrix proof to account explicitly for singular matrices and the absence of trace normalization.

Proof. Use Lemma 5.2 with $r = k$ and $B = H^{1/2}$ to choose $XX^\top = G$ with $\langle B, X \rangle = \sqrt{F_H(G)}$. For every $Z \succ 0$, the Frobenius Cauchy–Schwarz inequality gives

$$F_H(G) = \langle Z^{-1/2}B, Z^{1/2}X \rangle^2 \leq \text{tr}(GZ) \text{tr}(HZ^{-1}).$$

If $G, H \succ 0$, put $K = (G^{1/2}HG^{1/2})^{1/2}$ and $Z = G^{-1/2}KG^{-1/2}$. Then $ZGZ = H$, and $\text{tr}(GZ) = \text{tr}(HZ^{-1}) = \text{tr } K$, so equality holds.

For general positive semidefinite G, H , apply this construction to $G_\epsilon = G + \epsilon I$ and $H_\epsilon = H + \epsilon I$, obtaining $Z_\epsilon \succ 0$. Positivity gives

$$\inf_{Z \succ 0} \text{tr}(GZ) \text{tr}(HZ^{-1}) \leq \text{tr}(GZ_\epsilon) \text{tr}(HZ_\epsilon^{-1}) \leq F_{H_\epsilon}(G_\epsilon).$$

Continuity of positive semidefinite matrix square roots lets $\epsilon \downarrow 0$ and proves the reverse inequality. For fixed H , each expression in the infimum is linear in G ; an infimum of linear functions is concave. $\square$

Proof of Theorem 5.1. Suppose $r \geq k$ and $(k, r) \neq (1, 1)$. Given $G \succeq 0$ and $T \geq 0$, take $t = \sqrt{T}$. Lemma 5.2 says that one can satisfy $\langle B, X \rangle = -\beta t$ with $XX^\top = G$ if and only if $|\beta|\sqrt{T} \leq \sqrt{F_{BB^\top}(G)}$. This proves (5.2). Its right-hand side is convex by Lemma 5.3, proving HHC. When $k = r = 1$, every hyperplane in the two-dimensional domain is a line. Its quadratic image is a ray, hence convex. In that case the image formula need not hold: $x + t = 0$ has image $\{(G, T) : G = T \geq 0\}$, rather than $0 \leq T \leq G$.

Conversely, if $r < k$, the image of the hyperplane $t = 0$ is $\{(G, 0) : G \succeq 0, \text{rank } G \leq r\}$. The matrices $G_1 = \sum_{j=1}^r e_j e_j^\top$ and $G_2 = e_{r+1} e_{r+1}^\top$ belong to this image, but their midpoint has rank $r + 1$. The image is not convex. $\square$

Corollary 5.4 (Repeated quadratic blocks). *If $r \geq k$, then any finite family*

$$q_i(X, t) = \text{tr}(A_i X X^\top) + c_i t^2, \quad A_i \in \mathbb{S}^k, \quad c_i \in \mathbb{R},$$

has HHC, with no restriction on the number of forms.

Proof. On every hyperplane the image of $(q_1, \dots, q_m)$ is a linear image of the convex set $\Phi_{k,r}(L)$. $\square$

The necessity of $r \geq k$ concerns the *full* Gram map; a smaller collection of its output coordinates can have HHC below this threshold. Nor does the corollary allow arbitrary mixed terms $2t\langle D_i, X \rangle$. For $k = 2$ the image formula simplifies to

$$F_H(G) = \text{tr}(HG) + 2\sqrt{\det H \det G},$$

by expanding the square of the sum of two singular values. This two-row case yields the four-variable construction below.

The classical ingredients are the singular-value extremum and squared-fidelity concavity. The structural conclusion here is their sharp HHC formulation for a Gram map with one additional scalar square. We do not assert a new fidelity inequality or a new general principle of semidefinite exactness. Results on quadratic matrix programming and rotation images provide related convexity mechanisms [2, 20]. In particular, Beck [3, Theorems 3.1 and 3.4] proves global image convexity for real quadratic matrix functions when their number is at most the replication count, and for one additional function under a positive definite combination of the quadratic blocks and a row dimension of at least two. These are global image results; an arbitrary hyperplane restriction generally destroys that repeated-block structure. Theorem 5.1 instead preserves every Gram entry and the scalar square while imposing one arbitrary real linear equation. The next section explains the SDP comparison relevant to its application.

6 HHC can require infinitely many quadratic inequalities

Fix $r \geq 2$ and let

$$\begin{aligned} f_1(u, v) &= \|u\|^2 - 1, & f_2(u, v) &= \|v\|^2 - 1, \\ f_3(u, v) &= \frac{1}{2} - u^\top v, & S_r &= \{(u, v) \in \mathbb{R}^r \times \mathbb{R}^r : f_1, f_2, f_3 < 0\}. \end{aligned} \quad (6.1)$$

Write $C_r = \text{conv } S_r$ and $D_r = \text{cl } C_r$, and let T_r be the corresponding system with all three inequalities nonstrict. The set S_r is nonempty, since $u = v = \sqrt{3/4}e_1$ is strictly feasible, and bounded by the two unit-ball constraints. In particular, C_r is nonempty and proper.

Proposition 6.1. *The homogenization of (6.1) has HHC. Its good aggregation multipliers are exactly $\mathcal{K}_r \setminus \{0\}$, where*

$$\mathcal{K}_r = \{\lambda \in \mathbb{R}_+^3 : \lambda_3^2 \leq 4\lambda_1\lambda_2\}. \quad (6.2)$$

Proof. Apply Corollary 5.4 with $k = 2$ and $X = \begin{pmatrix} u^\top \\ v^\top \end{pmatrix}$. The homogeneous map is the linear image $(G, T) \mapsto (G_{11} - T, G_{22} - T, T/2 - G_{12})$ of $\Phi_{2,r}$, so it has HHC.

Up to a coordinate permutation, the aggregated homogeneous matrix is

$$Q_\lambda = \left[\begin{pmatrix} \lambda_1 & -\lambda_3/2 \\ -\lambda_3/2 & \lambda_2 \end{pmatrix} \otimes I_r \right] \oplus [\lambda_3/2 - \lambda_1 - \lambda_2]. \quad (6.3)$$

If its two-by-two block has a negative eigenvalue, that eigenvalue repeats $r \geq 2$ times; the aggregation cannot be good. Otherwise the block is positive semidefinite, which for $\lambda \geq 0$ is equivalent to (6.2). For a nonzero such multiplier,

$$\lambda_3 \leq 2\sqrt{\lambda_1\lambda_2} \leq \lambda_1 + \lambda_2, \quad \lambda_3/2 - \lambda_1 - \lambda_2 < 0.$$

Thus Q_λ has exactly one negative eigenvalue. Moreover, f_λ is convex and strictly negative on S_r ; convexity makes it strictly negative on every finite convex combination of points in S_r . Both requirements for goodness hold. $\square$

6.1 A continuum of indispensable strict aggregations

Theorem 6.2. *Every exact description*

$$C_r = \bigcap_{\lambda \in \Lambda} \{(u, v) : f_\lambda(u, v) < 0\}, \quad \Lambda \subseteq \mathcal{K}_r \setminus \{0\},$$

contains a positive multiple of $(\tau, 1/\tau, 2)$ for every $\tau \in [1, 2]$. Hence such a description requires uncountably many distinct rays. No finite family of nonstrict good aggregations describes D_r .

Proof. Set $\epsilon = 1/10$. For $\tau \in [1, 2]$, choose u_τ, v_τ with Gram matrix

$$G_\tau = \begin{pmatrix} 1 - \epsilon/\tau & 2/5 \\ 2/5 & 1 - \epsilon\tau \end{pmatrix}. \quad (6.4)$$

Both diagonal entries are at least $4/5$, and $\det G_\tau \geq (4/5)^2 - (2/5)^2 = 12/25 > 0$, so these vectors exist in $\mathbb{R}^2$ and hence in $\mathbb{R}^r$. At $x_\tau = (u_\tau, v_\tau)$,

$$f(x_\tau) = \epsilon(-1/\tau, -\tau, 1), \quad f_\lambda(x_\tau) = \epsilon(-\lambda_1/\tau - \lambda_2\tau + \lambda_3) \leq 0. \quad (6.5)$$

The inequality follows from $\lambda_1/\tau + \lambda_2\tau \geq 2\sqrt{\lambda_1\lambda_2} \geq \lambda_3$. For nonzero $\lambda \in \mathcal{K}_r$, equality holds precisely when λ is a positive multiple of $\lambda(\tau) = (\tau, 1/\tau, 2)$. Indeed, equality forces both positive diagonal weights, $\lambda_1/\tau = \lambda_2\tau$, and $\lambda_3 = 2\sqrt{\lambda_1\lambda_2}$.

The strict valid aggregation $\lambda(\tau)$ excludes x_τ from C_r . Every other good ray is strictly satisfied there. Any exact strict description must therefore include this ray, proving the first assertion.

Now fix a finite good family and choose a ray $\lambda(\tau)$ it omits. Decrease both off-diagonal entries of G_τ by a sufficiently small $\eta > 0$. Positive definiteness is preserved, and every member of the finite family remains strictly satisfied, by continuity of its value as a linear function of the Gram entries. The omitted aggregate now has value $2\eta > 0$. It is nonpositive on D_r , so the new Gram matrix realizes a point outside D_r that satisfies every member of the finite family. This proves the nonstrict assertion. $\square$

Proposition 6.1 and Theorem 6.2 give an affirmative answer to Conjecture 3.1 of Blekherman et al. [7], already in four original variables and with three inequalities. The indispensability proof does not use the general HHC aggregation theorem. That theorem does ensure that an exact description by all good aggregations exists; it will also give a convenient hull formula.

6.2 Ordinary and closed hulls, and a finite SDP lift

For the rest of this section, abbreviate

$$p = 1 - \|u\|^2, \quad q = 1 - \|v\|^2, \quad h = u^\top v.$$

Theorem 6.3. *For every $r \geq 2$,*

$$C_r = \{(u, v) : p > 0, q > 0, h + \sqrt{pq} > 1/2\}, \quad (6.6)$$

$$D_r = \text{conv } T_r = \{(u, v) : p \geq 0, q \geq 0, h + \sqrt{pq} \geq 1/2\}. \quad (6.7)$$

Define the affine matrix

$$M(u, v, \sigma) = \begin{pmatrix} 1 & \sigma & u^\top \\ \sigma & 1 & v^\top \\ u & v & I_r \end{pmatrix}. \quad (6.8)$$

Then the two hulls have the finite lifted descriptions

$$C_r = \{(u, v) : \exists \sigma > 1/2, M(u, v, \sigma) \succ 0\}, \quad (6.9)$$

$$D_r = \{(u, v) : \exists \sigma \geq 1/2, M(u, v, \sigma) \succeq 0\}. \quad (6.10)$$

Furthermore, the coordinate multipliers $(1, 0, 0), (0, 1, 0)$ and any countable dense family of $\lambda(\tau) = (\tau, 1/\tau, 2)$, $\tau > 0$, describe D_r by nonstrict aggregations.

Proof. The hypotheses of Blekherman et al. [7, Theorem 2.9] hold: $n = 2r \geq 4$, S_r is nonempty with a proper hull, and its homogenization has HHC. Therefore C_r is the intersection of the strict aggregations in (6.2).

The cone is generated by the two coordinate rays and $\lambda(\tau)$ for $\tau > 0$. To check this directly, suppose $\lambda_3 > 0$ and put $t = \lambda_3/2$. The cone inequality allows a choice $t/\lambda_2 \leq \tau \leq \lambda_1/t$. Then

$$\lambda = t(\tau, 1/\tau, 2) + (\lambda_1 - t\tau)(1, 0, 0) + (\lambda_2 - t/\tau)(0, 1, 0),$$

with nonnegative coefficients. If $\lambda_3 = 0$, only the coordinate rays are needed. Thus the strict intersection is given by $p > 0, q > 0$, and

$$\tau p + q/\tau > 1 - 2h \quad (\tau > 0). \quad (6.11)$$

For $p, q > 0$, the left side attains its minimum $2\sqrt{pq}$ at $\tau = \sqrt{q/p}$. This proves (6.6).

The Schur complement of I_r in (6.8) is

$$\begin{pmatrix} p & \sigma - h \\ \sigma - h & q \end{pmatrix}.$$

Consequently the projection in (6.10) is exactly the set on the right of (6.7): a feasible $\sigma \geq 1/2$ exists if and only if $h + \sqrt{pq} \geq 1/2$. The same calculation with strict positive definiteness and $\sigma > 1/2$ gives (6.9). This uses that the interval $(h - \sqrt{pq}, h + \sqrt{pq})$ intersects $(1/2, \infty)$ exactly when its upper endpoint exceeds $1/2$.

Let L_r denote the closed set on the right of (6.7). It contains C_r , so $D_r \subseteq L_r$. For $z = (u, v) \in L_r$, choose a feasible lift σ and mix it with the strictly positive definite matrix $M(0, 0, 1/2)$:

$$(1 - \theta)M(u, v, \sigma) + \theta M(0, 0, 1/2) = M((1 - \theta)u, (1 - \theta)v, (1 - \theta)\sigma + \theta/2) \succ 0$$

for $0 < \theta < 1$. Its scalar entry is at least $1/2$; if it equals $1/2$, positive definiteness permits a sufficiently small increase. Formula (6.9) then puts $(1 - \theta)z$ in C_r . Sending $\theta \downarrow 0$ proves $L_r \subseteq D_r$.

The original closed set T_r is compact, contains S_r , and is contained in $L_r = D_r$. Its convex hull is compact and hence closed, so $D_r \subseteq \text{conv } T_r \subseteq D_r$. This proves the asserted equality for T_r without assuming that closure commutes with intersecting the original constraints.

Finally, the coordinate inequalities give $p, q \geq 0$. If the nonstrict version of (6.11) holds on a dense set of positive τ , continuity extends it to all $\tau > 0$. Its infimum is $2\sqrt{pq}$, also when $p = 0$ or $q = 0$ by taking a limit at an endpoint. The resulting condition is exactly (6.7). $\square$

For $r \geq 3$, there is also a direct sufficiency proof of (6.6). If the point is not already in S_r , choose a unit vector e perpendicular to both u, v . For $\alpha < 1$ sufficiently close to one, the two points

$$(u + \alpha\sqrt{p}e, v + \alpha\sqrt{q}e), \quad (u - \alpha\sqrt{p}e, v - \alpha\sqrt{q}e)$$

belong to S_r and have midpoint (u, v) : their squared norms are $1 - (1 - \alpha^2)p$ and $1 - (1 - \alpha^2)q$, and their inner product is $h + \alpha^2\sqrt{pq} > 1/2$. Necessity follows directly from convex combinations as

well. For a finite random vector (U, V) supported in S_r with mean (u, v) , put $a = \mathbb{E}\|U - u\|^2 < p$ and $b = \mathbb{E}\|V - v\|^2 < q$. Cauchy–Schwarz gives

$$1/2 < \mathbb{E}(U^\top V) = h + \mathbb{E}[(U - u)^\top (V - v)] \leq h + \sqrt{ab} < h + \sqrt{pq}.$$

The perpendicular-direction construction requires $r \geq 3$; the HHC proof above establishes the formula also at $r = 2$.

6.3 Cardinality limits for quadratic descriptions in the original variables

The preceding obstruction concerned good aggregations. Allowing arbitrary quadratics still requires uncountably many strict inequalities for the open hull and infinitely many nonstrict inequalities for the closed hull.

Theorem 6.4. *For $r \geq 2$, no countable conjunction of strict quadratic inequalities in (u, v) describes C_r . No finite conjunction of nonstrict quadratic inequalities in (u, v) describes D_r . In both statements the quadratics may be nonconvex and need not be aggregations of the original three inequalities.*

Proof. Restrict to the plane $u = xe_1$, $v = ye_2$. By Theorem 6.3, the planar sections are

$$\{(x, y) : x^2 < 1, y^2 < 1, P(x, y) > 0\}, \quad \{(x, y) : x^2 \leq 1, y^2 \leq 1, P(x, y) \geq 0\},$$

respectively, where

$$P(x, y) = (1 - x^2)(1 - y^2) - 1/4. \tag{6.12}$$

Both have the boundary arc

$$\Gamma = \{(x, \psi(x)) : 1/4 \leq x \leq 1/2\}, \quad \psi(x) = \sqrt{\frac{3/4 - x^2}{1 - x^2}}.$$

The radicand is positive on a neighborhood of this interval and is strictly below one. Hence Γ is an analytic arc inside the two ball bounds, with feasible and infeasible points in every planar neighborhood.

We first show that a nonzero polynomial $q(x, y)$ of total degree at most two can vanish at only finitely many points of Γ . The rational function

$$R(x) = \frac{3/4 - x^2}{1 - x^2}$$

is not a square in $\mathbb{R}(x)$, since its zeros at $\pm\sqrt{3}/2$ and poles at ± 1 are simple. Thus $y^2 - R(x)$ is irreducible in $\mathbb{R}(x)[y]$. Divide q by this polynomial in y ; the remainder has the form $a(x)y + b(x)$ with $a, b \in \mathbb{R}(x)$. If $q(x, \psi(x))$ vanished identically on a subinterval and $a \neq 0$, then $\psi = -b/a$ there, forcing $R = (b/a)^2$ as a rational identity, a contradiction. If $a = 0$, then $b = 0$ on that subinterval and hence identically. In the remaining case $a = b = 0$, the polynomial $y^2 - R$ divides q in $\mathbb{R}(x)[y]$.

Viewed as a polynomial in y over $\mathbb{R}[x]$, P is primitive: its two nonzero coefficients $x^2 - 1$ and $3/4 - x^2$ are coprime. Gauss’s lemma therefore makes P irreducible in $\mathbb{R}[x, y]$ and transfers that divisibility to $P \mid q$ in $\mathbb{R}[x, y]$. This is impossible for nonzero q of degree at most two, because P has total degree four. We have shown that $q(x, \psi(x))$ cannot vanish identically on a subinterval. It is real analytic on a neighborhood of the compact parameter interval; infinitely many zeros there would have an accumulation point and force such identical vanishing. Its zeros on Γ are therefore finite.

Suppose a countable family of strict quadratic inequalities described C_r , and restrict their polynomials to the plane. None is identically zero, since the planar origin belongs to C_r and all its inequalities must be strict there. At a point of Γ , continuity from the planar feasible region makes all restricted polynomials nonpositive. At least one must vanish, since otherwise the strict description would include this boundary point. Thus a countable union of finite zero sets would cover the uncountable arc Γ , a contradiction.

For a hypothetical finite nonstrict description of D_r , discard the restricted polynomials that vanish identically on the plane. Every remaining polynomial is nonpositive at each point of Γ . At least one must vanish: if they were all strictly negative, their finite conjunction would contain a planar neighborhood, including points outside D_r . Finitely many finite zero sets cannot cover Γ , giving a contradiction. The argument also covers the case that no nonzero restrictions remain, since then the description would include the entire plane. $\square$

Theorem 6.4 concerns conjunctions in the original variables. The finite semidefinite lifts in Theorem 6.3 remain available. Every exact strict quadratic description in the original variables needs uncountably many inequalities. For the closed hull, the obstruction excludes finite descriptions only; a countable dense family of nonstrict good aggregations suffices. These cardinality statements do not give a runtime lower bound for optimization.

6.4 Relation to previous infinite examples and finite bounds

Infinite necessity itself predates the HHC framework. Dey et al. [12, Proposition 2.8 and Section 7.4] give a closed two-variable example with defining functions

$$g_1 = x^2 - 1, \quad g_2 = y^2 - 1, \quad g_3 = -(x - 1)^2 - (y - 1)^2 + 1.$$

It fails HHC: on the homogeneous hyperplane $y = 0$, the image is $(x^2 - t^2, -t^2, -x^2 + 2xt - t^2)$. The images of $(x, t) = (1, 0)$ and $(0, 1)$ have midpoint $(0, -1/2, -1)$, which would require $x^2 = t^2 = 1/2$ and $xt = 0$. Moreover, their displayed multipliers $(a^2, (1 - a)^2, a^2 - a + 1)$, $0 < a < 1$, have leading matrix $\text{diag}(a - 1, -a)$ and therefore at least two negative homogeneous eigenvalues. That family is outside the good-aggregation class used here. Another infinite necessity result, Dey et al. [13, Theorem 2], aggregates boxed bilinear equalities with signed multipliers and convexifies each aggregated set before intersecting. It concerns a different aggregation operation.

The relevant distinction is therefore precise: the construction in (6.1) has HHC and still requires a continuum of good rays for its strict hull, with the stronger original-space obstruction in Theorem 6.4. To the best of our knowledge, this gives an explicit resolution of the HHC finiteness question in Blekherman et al. [7, Conjecture 3.1]. We do not claim that infinite aggregation in general, or a continuously varying family of active inequalities, first arises here.

Semidefinite convexification is also established prior theory. Quadratic matrix programming was studied by Beck [2]. The convex-hull result of Wang and Kılınç-Karzan [25, Section 4.1] applies to (6.1) for $r \geq 3$: take zero objective, three constraints, replication count r , and the positive definite sum $A_1 + A_2 = I_{2r}$ to meet their definiteness assumption. Their paper appeared as [24]. This comparison concerns the closed feasible-set hull, obtained from the zero-objective epigraph, rather than an automatic replacement of nonstrict inequalities by strict ones. Brun et al. [11, Section 3.2.1, Corollary 1] also derive a related inner-product hypograph hull over two balls. Fixing the level after convexification does not by itself prove the fixed-level hull here. The explicit proof above covers $r = 2$ and both choices of strictness; its finite lift is not claimed as a new SDP convexification principle.

Finally, the example is outside the homogeneous definiteness assumptions of known finiteness results. For any signed λ , positive definiteness of the two-by-two block in (6.3) implies $\lambda_1 + \lambda_2 > |\lambda_3|$, whereas positivity of the scalar block requires $\lambda_3/2 > \lambda_1 + \lambda_2$. These are incompatible. Thus the Q_i have no positive definite real linear combination, even though $A_1 + A_2 \succ 0$. The spectral determinant is

$$\det Q_\lambda = (\lambda_1 \lambda_2 - \lambda_3^2/4)^r (\lambda_3/2 - \lambda_1 - \lambda_2),$$

which has a repeated factor. The common real projective zero set is nonempty: take $t = 1$, $u = e_1$, and $v = e_1/2 + \sqrt{3}e_2/2$. These facts prevent an application of finiteness results requiring homogeneous PDL, a smooth spectral curve, or absence of common projective zeros. In particular, the four-aggregation result in the inspected version of Blekherman and Dunbar [5, Theorem 1.4], published as [6], retains additional hypotheses. It does not contradict the construction here.

7 Finite approximation and one-objective exactness

The continuum in Theorem 6.2 is necessary for an exact strict description. For approximate descriptions of the closed hull, a finite family suffices, and its required size can be quantified. Throughout this section $r \geq 2$, and D_r and $\mathcal{K}_r$ retain their meanings from Section 6. For a finite family $F \subset \mathcal{K}_r \setminus \{0\}$, put

$$P_F = \bigcap_{\lambda \in F} \{x \in \mathbb{R}^{2r} : f_\lambda(x) \leq 0\}, \quad e_N = \inf_{|F| \leq N} d_H(D_r, P_F).$$

Here d_H is Euclidean Hausdorff distance, with value $+\infty$ for unbounded P_F . Since $D_r \subseteq P_F$ and D_r is compact, $d_H(D_r, P_F) = \sup_{x \in P_F} \text{dist}(x, D_r)$, also with this extended value. Multipliers may be rescaled arbitrarily and may lie in the interior of the good cone. The definition counts inequalities; it does not assert that an optimal family attaining e_N exists.

Theorem 7.1. *For every integer $N \geq 2$ and every $r \geq 2$,*

$$\frac{\sqrt{2}}{2000N^2} \leq e_N \leq \frac{5\sqrt{2}\pi^2}{16(N-1)^2}. \quad (7.1)$$

Thus $e_N = \Theta(N^{-2})$, with constants independent of r . Equivalently, as $\varepsilon \downarrow 0$, a uniform Hausdorff error at most ε requires and suffices with $\Theta(\varepsilon^{-1/2})$ good aggregations.

The rate is specific to the good multipliers of this system. Although Theorem 6.4 permits arbitrary quadratic inequalities, the quantitative lower bound here does not. The constants in (7.1) are convenient explicit bounds; no optimality of their values is claimed.

More explicitly, put $c_0 = \sqrt{2}/2000$ and $B_0 = 5\sqrt{2}\pi^2/16$. Any family of at most $N \geq 1$ cuts whose error is at most $\varepsilon > 0$ must have $N \geq \sqrt{c_0/\varepsilon}$; the lower-bound proof below also applies when $N = 1$. Conversely, the equally spaced family with $\lceil \sqrt{B_0/\varepsilon} \rceil + 1$ cuts suffices, and its size is less than $\sqrt{B_0/\varepsilon} + 2$. This sufficiency uses an explicit construction, not attainment of the infimum defining e_N .

7.1 Angle sampling and a uniform radial correction

For $x = (u, v)$ define

$$B(x) = \begin{pmatrix} 1 - \|u\|^2 & u^\top v - 1/2 \\ u^\top v - 1/2 & 1 - \|v\|^2 \end{pmatrix}, \quad q(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \lambda(\theta) = (\cos^2 \theta, \sin^2 \theta, 2 \sin \theta \cos \theta).$$

The use of $\lambda(\theta)$ here is an angle parametrization of the rays $\lambda(\tau)$ in Section 6: for $0 < \theta < \pi/2$ the rays agree when $\tau = \cot \theta$. The endpoints give the two coordinate rays. These rays generate $\mathcal{K}_r$, so Theorem 6.3 gives

$$x \in D_r \iff h_x(\theta) := q(\theta)^\top B(x)q(\theta) = -f_{\lambda(\theta)}(x) \geq 0 \quad (0 \leq \theta \leq \pi/2). \quad (7.2)$$

Only nonnegative directions $q(\theta)$ are tested: $B(x)$ need not be positive semidefinite for $x \in D_r$.

Lemma 7.2. *Let $0 = \theta_1 < \dots < \theta_N = \pi/2$, and let $\Delta = \max_j(\theta_{j+1} - \theta_j)$. The relaxation P_F using $F = \{\lambda(\theta_j) : 1 \leq j \leq N\}$ satisfies*

$$d_H(D_r, P_F) \leq \frac{5\sqrt{2}}{4} \Delta^2.$$

Proof. The endpoint cuts give $\|u\|, \|v\| \leq 1$ on P_F . On this product of balls, the diagonal entries of $B(x)$ lie in $[0, 1]$ and its off-diagonal entry lies in $[-3/2, 1/2]$. The maximum absolute row sum therefore gives $\|B(x)\|_{\text{op}} \leq 5/2$. Since $q'' = -q$,

$$h_x''(\theta) = 2q'(\theta)^\top B(x)q'(\theta) - 2q(\theta)^\top B(x)q(\theta), \quad |h_x''(\theta)| \leq 10.$$

If a minimizing angle is an endpoint, the minimum is nonnegative. Otherwise its first derivative is zero, and a mesh point lies within $\Delta/2$. Taylor's upper bound at that mesh point, whose value is nonnegative, gives

$$h_x(\theta) \geq -a \quad \text{for all } \theta, \quad a = 5\Delta^2/4. \quad (7.3)$$

This estimate holds at every point of P_F .

$$\text{Now } B(0) = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix} \succeq I_2/2 \text{ and}$$

$$B(sx) = s^2 B(x) + (1 - s^2) B(0) \quad (0 \leq s \leq 1).$$

Choose $s = (1 + 2a)^{-1/2}$. Equations (7.2) and (7.3) imply $h_{sx}(\theta) \geq -s^2 a + (1 - s^2)/2 = 0$, so $sx \in D_r$. Since $\|x\| \leq \sqrt{2}$,

$$\text{dist}(x, D_r) \leq \|x - sx\| \leq \sqrt{2}(1 - (1 + 2a)^{-1/2}) \leq \sqrt{2}a.$$

The last inequality follows from the tangent bound $(1 + 2a)^{-1/2} \geq 1 - a$ for the convex function on $a \geq 0$. Taking the supremum proves the claim. $\square$

Taking N equally spaced angles gives $\Delta = \pi/[2(N - 1)]$ and proves the upper bound in Theorem 7.1, including $N = 2$. Exact rational coefficients give the same asymptotic rate.

Corollary 7.3. *For $N \geq 3$ set $m = \lfloor (N - 1)/2 \rfloor$. The distinct multipliers*

$$(m^2, j^2, 2mj), \quad (j^2, m^2, 2mj), \quad j = 0, \dots, m, \quad (7.4)$$

define a relaxation using $2m + 1 \leq N$ cuts with

$$d_H(D_r, P_F) \leq \frac{5\sqrt{2}}{4m^2}.$$

Every coefficient is an integer between 0 and $2m^2$ and therefore requires at most $2\lceil \log_2 N \rceil + 3$ binary digits, hence $O(\log N)$ bits.

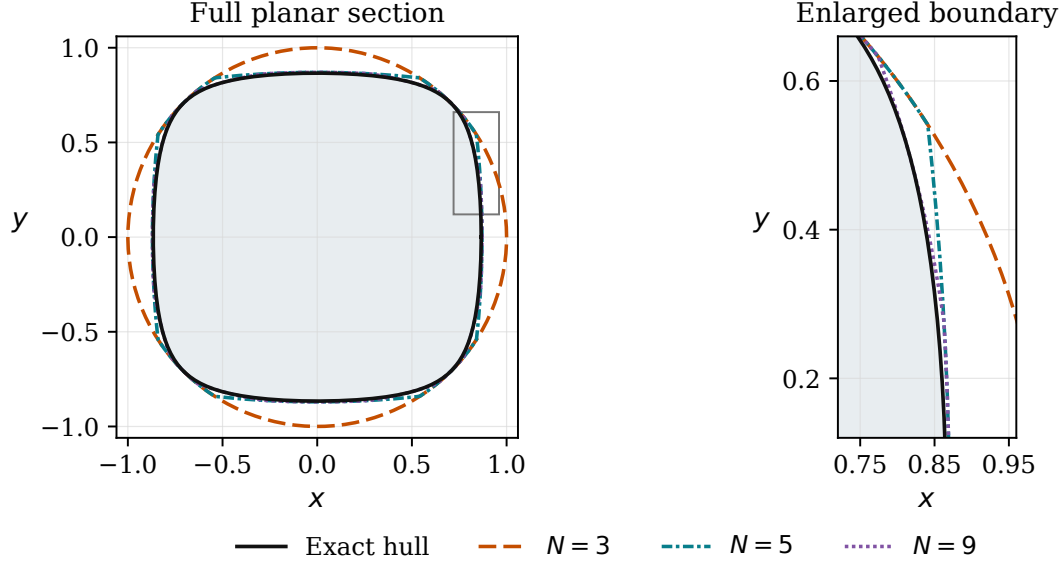


Figure 1: Sections by the plane $u = xe_1$, $v = ye_2$. The exact closed hull (black) is $(1 - x^2)(1 - y^2) \geq 1/4$ inside $[-1, 1]^2$. Colored curves bound the outer approximations from $N = 3, 5, 9$ equally spaced angles in Lemma 7.2; the right panel enlarges part of the first quadrant. The same planar sections occur for every $r \geq 2$. The curves are computed from explicit radial formulas and illustrate the approximations; they are not estimates of their full Hausdorff errors.

Proof. The two lists share only the middle ray, and each multiplier satisfies $\lambda_3^2 = 4\lambda_1\lambda_2$ exactly. Their angles are $\arctan(j/m)$ and their reflections about $\pi/4$. They include both endpoints and have gaps at most $1/m$, because $\arctan$ is 1-Lipschitz on $[0, 1]$. Apply Lemma 7.2. For the digit count, $N < 2^{\lfloor \log_2 N \rfloor + 1}$ gives $2m^2 \leq 2N^2 < 2^{2\lfloor \log_2 N \rfloor + 3}$; the bound also holds for the coefficient zero, encoded by one digit. $\square$

The bit bound concerns the three multiplier coefficients, not the total encoding of a quadratic in $2r$ variables or numerical solution error. Preserving the rank-one identity in (7.4) is useful: independently rounding its three entries need not preserve goodness.

7.2 A lower bound allowing interior multipliers

Proof of the lower bound in Theorem 7.1. Fix $F \subset \mathcal{K}_r \setminus \{0\}$ with $|F| \leq N$, and put

$$\eta = \frac{1}{400N^2}, \quad \epsilon = 1/10.$$

Use the $N + 1$ parameters $\tau_j = 1 + j/N$, $j = 0, \dots, N$. For each such τ , choose $x_{\tau, \eta} = (u_{\tau, \eta}, v_{\tau, \eta})$ with Gram matrix

$$G_{\tau, \eta} = \begin{pmatrix} 1 - \epsilon/\tau & 2/5 - \eta \\ 2/5 - \eta & 1 - \epsilon\tau \end{pmatrix}. \quad (7.5)$$

Here $0 < \eta < 1/10$, both diagonal entries are at least $4/5$, and the off-diagonal entry has absolute value at most $2/5$. Thus $\det G_{\tau, \eta} \geq 12/25 > 0$. Such vectors exist in $\mathbb{R}^2$ and embed in $\mathbb{R}^r$, with both norms at most one. They satisfy

$$f(x_{\tau, \eta}) = (-\epsilon/\tau, -\epsilon\tau, \epsilon + \eta). \quad (7.6)$$

Consider any $\lambda \in F$. If either λ_1 or λ_2 vanishes, goodness forces $\lambda_3 = 0$, and the nonzero cut is strictly satisfied at every witness. Otherwise put $t = \sqrt{\lambda_1/\lambda_2} > 0$ and $b = \sqrt{\lambda_1\lambda_2}$. If the cut excludes a witness, then (7.6) gives

$$\epsilon(\lambda_1/\tau + \lambda_2\tau - \lambda_3) < \eta\lambda_3.$$

Using $\lambda_3 \leq 2b$ on both sides and dividing by ϵb yields $t/\tau + \tau/t < 2a$, where $a = 1 + 10\eta$. If the same cut excludes a second parameter σ , adding the two inequalities and applying the arithmetic–geometric mean inequality gives

$$4a > (\tau + \sigma) \left(\frac{t}{\tau\sigma} + \frac{1}{t} \right) \geq \frac{2(\tau + \sigma)}{\sqrt{\tau\sigma}}.$$

It follows that

$$(\tau - \sigma)^2 < 4\tau\sigma(a^2 - 1) \leq 16(a^2 - 1) = \frac{4}{5N^2} + \frac{1}{100N^4} < \frac{1}{N^2}. \quad (7.7)$$

Distinct grid parameters differ by at least $1/N$. Thus one cut excludes at most one of the $N + 1$ witnesses. At most N cuts leave some $x_{\tau,\eta}$ in P_F . This finite pigeonhole argument imposes no restriction on t or extremality of λ .

For this witness the good multiplier $\lambda_* = (\tau, 1/\tau, 2)$ has $f_{\lambda_*}(x_{\tau,\eta}) = 2\eta > 0$. Its leading matrix is

$$\begin{pmatrix} \tau & -1 \\ -1 & 1/\tau \end{pmatrix} \otimes I_r,$$

which is positive semidefinite with operator norm $\tau + 1/\tau \leq 5/2$. Hence $\|\nabla f_{\lambda_*}(x)\| \leq 5\sqrt{2}$ on the convex product of the two unit balls. Both the witness and every $y \in D_r$ lie in that product, and $f_{\lambda_*}(y) \leq 0$. The mean-value inequality therefore gives

$$2\eta \leq f_{\lambda_*}(x_{\tau,\eta}) - f_{\lambda_*}(y) \leq 5\sqrt{2}\|x_{\tau,\eta} - y\|.$$

Consequently $d_H(D_r, P_F) \geq \sqrt{2}\eta/5$, proving the stated lower bound. For an unbounded relaxation the bound holds with infinite left side. $\square$

7.3 Uniform objective gaps and an exact cut for one objective

For a compact convex set L , write $h_L(c) = \max_{x \in L} c^\top x$. If P_F is bounded, then

$$d_H(D_r, P_F) = \sup_{\|c\|=1} (h_{P_F}(c) - h_{D_r}(c)). \quad (7.8)$$

Indeed, choosing a closest point in D_r bounds each support gap above by the Hausdorff distance. Conversely, for $x \in P_F \setminus D_r$ and its projection y onto D_r , the unit vector $c = (x - y)/\|x - y\|$ satisfies $h_{D_r}(c) = c^\top y$ and $h_{P_F}(c) \geq c^\top x$. Taking the supremum gives the reverse inequality; the case $P_F = D_r$ is immediate.

Thus Theorem 7.1 quantifies the worst gap over unit linear objectives for one common bounded relaxation. It applies to the final family even if the cuts were selected adaptively. It does not give an iteration lower bound for solving one prescribed objective. The next proposition makes that distinction explicit.

Proposition 7.4. *For every nonzero $c \in \mathbb{R}^{2r}$, there is a nonzero good multiplier $\lambda \in \mathcal{K}_r$ such that*

$$\min_{x \in D_r} c^\top x = \min\{c^\top x : f_\lambda(x) \leq 0\}. \quad (7.9)$$

Both minima are attained at a common point.

Proof. Let $F(x) = (f_1(x), f_2(x), f_3(x))$ and $\mathcal{K}_r^* = \{z : \lambda^\top z \geq 0 \text{ for all } \lambda \in \mathcal{K}_r\}$. Convexity of every f_λ for $\lambda \in \mathcal{K}_r$ implies

$$\theta F(x) + (1 - \theta)F(y) - F(\theta x + (1 - \theta)y) \in \mathcal{K}_r^* \quad (0 \leq \theta \leq 1).$$

It follows directly that

$$E = \{(F(x) + z, c^\top x + t) : x \in \mathbb{R}^{2r}, z \in \mathcal{K}_r^*, t \geq 0\}$$

is convex. It has nonempty interior because $\mathcal{K}_r^*$ contains $\mathbb{R}_+^3$, and t may increase freely. The closed good-aggregation description of D_r says that $x \in D_r$ exactly when $-F(x) \in \mathcal{K}_r^*$. Let $v = \min_{D_r} c^\top x$ and choose a minimizer x_* , which exists by compactness. Then $(0, v) \in E$, whereas $(0, v - \delta) \notin E$ for every $\delta > 0$. Thus $(0, v)$ is a boundary point. A supporting hyperplane gives a nonzero pair (λ, μ) satisfying

$$\lambda^\top (F(x) + z) + \mu(c^\top x + t) \geq \mu v \quad (x \in \mathbb{R}^{2r}, z \in \mathcal{K}_r^*, t \geq 0).$$

This supporting-hyperplane assertion does not require E to be closed: its nonempty interior has the same interior as its closure, so one may support that closure at the included boundary point. The recession directions z, t and the bipolar theorem force $\lambda \in \mathcal{K}_r$ and $\mu \geq 0$.

For every nonzero $\lambda \in \mathcal{K}_r$,

$$f_\lambda(0) = -\lambda_1 - \lambda_2 + \lambda_3/2 \leq -(\lambda_1 + \lambda_2)/2 < 0.$$

Evaluation at zero therefore rules out $\mu = 0$. Normalize $\mu = 1$ to obtain $f_\lambda(x) + c^\top x \geq v$ for all x . Evaluation at x_* , where $f_\lambda(x_*) \leq 0$, gives $f_\lambda(x_*) = 0$. Also $\lambda \neq 0$, since otherwise a nonzero linear function would be bounded below on all of $\mathbb{R}^{2r}$. For every x satisfying $f_\lambda(x) \leq 0$ we now have $c^\top x \geq v - f_\lambda(x) \geq v$, while x_* attains equality. Proposition 6.1 ensures goodness. $\square$

This is the usual strict-feasibility argument for conic convex duality, specialized to the cone of convex aggregations; see Boyd and Vandenberghe [9, Section 5.9]. It is not a new duality principle. The strict point used is the origin for the convex hull description, not for the original three inequalities. Finding the objective-dependent multiplier can itself require solving the original convex problem. The proposition gives no bound on that cost.

7.4 Comparison with approximation theory

The exponent two is familiar in convex approximation. Rote [21] proves optimal $O(N^{-2})$ error bounds for piecewise linear upper and lower approximations of univariate convex functions on compact intervals with finite endpoint slopes, using function values and supporting slopes, and treats planar convex figures as well. This is particularly close context for sampling a one-dimensional parameter family. His tangent-and-chord approximations are different from the prescribed quadratic sublevel sets considered here. The classical polyhedral bounds of Bronshteyn and Ivanov [10], and the modern area-sensitive bounds and comparison in Arya et al. [1, Section 1 and Theorems 1–2], concern approximation by vertices, linear halfspaces, or piecewise affine functions. They provide the appropriate context for quadratic local error from sampling a one-dimensional family; they do not directly give Theorem 7.1 for this prescribed cone of quadratic cuts in $2r$ dimensions.

The specific addition is a two-sided bound for the HHC example, including a lower bound against arbitrary interior good multipliers, constants uniform in the replication dimension, and an exact rational construction. It quantifies the size of a finite collection of good aggregations as an

alternative to the exact finite SDP lift in Theorem 6.3. No new general approximation exponent, optimal constant, numerical conditioning guarantee, solver improvement, or runtime lower bound is asserted. Proposition 7.4 also prevents interpreting a uniform representation bound as a bound for one objective.

8 Four aggregations for three strict inequalities under PDLC

The preceding example needs infinitely many good aggregations and does not satisfy PDLC for its homogeneous matrices. We now establish the sharp finite bound for three strict inequalities under that condition. Here PDLC means that $Q_\theta \succ 0$ for some $\theta \in \mathbb{R}^3$; the coefficients of θ may have either sign.

Theorem 8.1 (Four aggregations for strict PDLC systems). *Let $n \geq 1$ and $m = 3$. Suppose that $S \neq \emptyset$, $\text{conv } S \neq \mathbb{R}^n$, and Q_1, Q_2, Q_3 satisfy PDLC. Then there are $1 \leq \ell \leq 4$ good multipliers $\lambda^1, \dots, \lambda^\ell$ such that*

$$\text{conv } S = \bigcap_{j=1}^{\ell} \{x : f_{\lambda^j}(x) < 0\}. \quad (8.1)$$

Each Q_{λ^j} has exactly one negative eigenvalue. The original matrices need not be linearly independent, and the original nonstrict feasible set need not be bounded or the closure of its interior. For every $n \geq 3$, four is necessary in some systems.

Blekherman and Dunbar’s four-bound for regular nonstrict systems [5, 6, Theorem 1.4] is the substantive external input. We transfer that bound to arbitrary strict systems by compact inward perturbation and a limit that preserves strict validity. Blekherman, Dey, and Sun previously gave an upper bound of six [7, Corollary 2.20]; their Example 2.21 supplies the sharpness construction below. Thus Theorem 8.1 rules out the six-necessary example proposed in their Conjecture 3.2. The number four and its lower-bound example are established antecedents; the extension here concerns the unrestricted strict setting.

Dunbar’s dissertation also states a four-bound for regular nonstrict sets without the no-points-at-infinity hypothesis [16, Theorem 5.0.5]. We acknowledge that stronger prior statement and use only the fully qualified published theorem below. To the best of our knowledge, the complete transfer to arbitrary strict systems in Theorem 8.1 has not previously been given. This claim concerns that transfer, including possible dependence of the original matrices, rather than the first statement of a four-bound without an infinity assumption. The negative-eigenvector limit used below has an antecedent for a fixed feasible set in the proof of Blekherman and Dunbar [5, Proposition 3.15, Section 8].

8.1 Regular inward perturbations

We state the external input with its standing independence assumption. For a triple $\tilde{Q}_i$ with leading blocks $\tilde{A}_i$, the phrase *no points at infinity* means

$$\{d \in \mathbb{R}^n : d^\top \tilde{A}_i d \leq 0 \text{ for all } i\} = \{0\}. \quad (8.2)$$

Theorem 8.2 (Blekherman–Dunbar, external input). *Let $\tilde{Q}_1, \tilde{Q}_2, \tilde{Q}_3 \in \mathbb{S}^{n+1}$ be linearly independent and satisfy PDLC, and set $C = \{x : (x, 1)^\top \tilde{Q}_i(x, 1) \leq 0 \text{ for all } i\}$. If $\text{int } C \neq \emptyset$, $C = \text{cl}(\text{int } C)$, and (8.2) holds, then*

$$\text{cl}(\text{conv } C) = \bigcap_{j=1}^{\ell} \left\{ x : (x, 1)^\top \left(\sum_i \mu_i^j \tilde{Q}_i \right) (x, 1) \leq 0 \right\} \quad (\ell \leq 4),$$

where each $\mu^j \geq 0$ is nonzero and each aggregated matrix has at most one negative eigenvalue.

This is Blekherman and Dunbar [5, 6, Theorem 1.4]; no smoothness assumption on the spectral curve is needed. The proof in the cited preprint explicitly includes $n = 1, 2$ in Proposition 8.7 and combines Propositions 8.6, 8.7, and 8.10. Consequently the input applies in every positive dimension used here.

Lemma 8.3 (Regular sublevels). *Let $h : M \rightarrow \mathbb{R}$ be continuous on a space with a countable base. For all but at most countably many $c \in \mathbb{R}$,*

$$\{h \leq c\} = \text{cl}_M\{h < c\}.$$

Proof. Continuity gives the inclusion from right to left. If it is strict, there is z with $h(z) = c$ and an open neighborhood U of z containing no point where $h < c$. Choose a basic open set B with $z \in B \subseteq U$. Then $c = \inf_{y \in B} h(y)$. There are at most countably many such real infima, one for each basic open set. $\square$

Lemma 8.4 (Compact inward systems). *Under the hypotheses of Theorem 8.1, there are positive definite matrices R_1, R_2, R_3 and a decreasing sequence $\varepsilon_k \downarrow 0$ such that, on defining*

$$\begin{aligned} Q_i^\varepsilon &= Q_i + \varepsilon R_i, & w_i(x) &= (x, 1)^\top R_i(x, 1), \\ S_\varepsilon &= \{x : f_i(x) + \varepsilon w_i(x) < 0 \ \forall i\}, & C_\varepsilon &= \{x : f_i(x) + \varepsilon w_i(x) \leq 0 \ \forall i\}, \end{aligned}$$

each selected triple satisfies every hypothesis of Theorem 8.2. Moreover, C_{ε_k} is compact, $C_{\varepsilon_k} = \text{cl } S_{\varepsilon_k}$, $S_{\varepsilon_k} \subseteq S$, and every fixed finite subset of S is contained in S_{ε_k} for all sufficiently large k .

Proof. Since $n + 1 \geq 2$, take, with standard basis vectors e_1, e_2 ,

$$R_1 = I, \quad R_2 = I + e_1 e_1^\top, \quad R_3 = I + (e_1 + e_2)(e_1 + e_2)^\top.$$

These are positive definite and linearly independent: in a vanishing linear combination the $(1, 2)$ entry first forces the coefficient of R_3 to vanish, and the two diagonal entries then force the others to vanish. Write $B_i \succ 0$ for their leading $n \times n$ blocks. Lemma 2.4 excludes a nonzero d satisfying $d^\top A_i d < 0$ for every i . Therefore, for every $\varepsilon > 0$, the inequalities $d^\top (A_i + \varepsilon B_i) d \leq 0$ for all i force $d = 0$. Thus the perturbed system has no points at infinity.

It is also bounded. Otherwise there would be $x_j \in C_\varepsilon$ with $\|x_j\| \rightarrow \infty$. After taking a subsequence, $x_j / \|x_j\| \rightarrow d$ with $\|d\| = 1$. Divide each perturbed inequality by $\|x_j\|^2$ and take limits to obtain the forbidden nonpositive leading inequalities at d . Closedness now makes C_ε compact.

Fix $x^0 \in S$ and a PDLIC certificate θ . For every sufficiently small $\varepsilon > 0$, $x^0 \in S_\varepsilon$ and $Q_\theta + \varepsilon \sum_i \theta_i R_i \succ 0$, by strictness and continuity. The perturbed matrices are independent except at finitely many parameters. Indeed, choose three coordinates of $\mathbb{S}^{n+1}$ on which the coordinate matrix of the R_i has nonzero determinant. The corresponding determinant for the $Q_i + \varepsilon R_i$ is a cubic polynomial whose leading coefficient is that nonzero determinant. Its zeros form a finite set. This argument does not assume independence of the original Q_i .

Because every w_i is positive, the continuous function

$$h(x) = \max_{1 \leq i \leq 3} \frac{f_i(x)}{w_i(x)}$$

satisfies $S_\varepsilon = \{h < -\varepsilon\}$ and $C_\varepsilon = \{h \leq -\varepsilon\}$. Apply Lemma 8.3 and choose $\varepsilon_k \downarrow 0$ so that $-\varepsilon_k$ avoids its countable exceptional levels and ε_k avoids the finite dependence set, with every parameter

small enough for strict feasibility and PDLC. Such a sequence exists because every nonempty real interval is uncountable. Then

$$C_{\varepsilon_k} = \text{cl } S_{\varepsilon_k} = \text{cl}(\text{int } C_{\varepsilon_k}), \quad \text{int } C_{\varepsilon_k} \neq \emptyset.$$

The second equality follows from $S_{\varepsilon_k} \subseteq \text{int } C_{\varepsilon_k} \subseteq C_{\varepsilon_k}$ and closedness. Positivity of w_i gives $S_{\varepsilon} \subseteq S$. Finally, finitely many strict inequalities at finitely many points remain strict for all sufficiently small ε , proving the last assertion. $\square$

8.2 Strictification and the multiplier limit

Two details are essential: a redundant nonstrict quadratic can remove interior points when made strict, and a limiting quadratic can lose strict validity unless its negative component is controlled.

Lemma 8.5 (Strictification). *Let $U \subset \mathbb{R}^n$ be nonempty and open with bounded hull. Suppose*

$$\text{cl}(\text{conv } U) = \bigcap_{j=1}^q \{x : p_j(x) \leq 0\},$$

where the p_j are quadratic polynomials. After deleting every p_j that is globally nonpositive, at least one inequality remains and

$$\text{conv } U = \bigcap_{j \text{ retained}} \{x : p_j(x) < 0\}.$$

Proof. Deleting such polynomials preserves the nonstrict intersection. Some inequality remains because a nonempty bounded hull is not $\mathbb{R}^n$. If a quadratic p vanishes at an interior point x of $\{p \leq 0\}$, then x is a local maximum with value zero. Its gradient vanishes and its Hessian is negative semidefinite. The exact quadratic Taylor formula therefore gives $p \leq 0$ everywhere. For every retained polynomial this is impossible, so $\text{int}\{p_j \leq 0\} = \{p_j < 0\}$. The interior of a finite intersection equals the intersection of its interiors. Finally, the nonempty open convex set $\text{conv } U$ equals the interior of its closure, which proves the formula. $\square$

For completeness, the relevant negative components have a direct conic description. If a symmetric matrix M has exactly one negative eigenvalue $-a < 0$, choose its unit eigenvector v and put $P = M + avv^\top$. Then $P \succeq 0$, $Pv = 0$, and

$$\{z : z^\top M z < 0\} = \{z : \|P^{1/2} z\| < \sqrt{a} v^\top z\} \cup \{z : \|P^{1/2} z\| < -\sqrt{a} v^\top z\}. \quad (8.3)$$

Both sets are convex, disjoint, and nonempty. This formula also covers singular M and the case $P = 0$. A connected subset of the negative sublevel lies in one of the two sets.

Proof of Theorem 8.1, upper bound. Use the sequence from Lemma 8.4 and abbreviate $S_k = S_{\varepsilon_k}$, $C_k = C_{\varepsilon_k}$. Because $C_k = \text{cl } S_k$,

$$\text{cl}(\text{conv } C_k) = \text{cl}(\text{conv } S_k).$$

Indeed, each closed convex set containing S_k also contains C_k , and the reverse inclusion follows from $S_k \subseteq C_k$. Theorem 8.2 followed by Lemma 8.5 gives a description of $\text{conv } S_k$ by between one and four strict aggregations of the perturbed inequalities. Every retained aggregation is good for S_k and has exactly one negative homogeneous eigenvalue, since it is strictly negative at x^0 . Duplicate multipliers to obtain four indices and normalize each in the simplex $\Delta_3 = \{\lambda \geq 0 : \sum_i \lambda_i = 1\}$.

Write them as $\lambda^{k,j}$, $1 \leq j \leq 4$. By compactness, after a common subsequence $\lambda^{k,j} \rightarrow \lambda^j \in \Delta_3$ for every j . Their matrices satisfy

$$M_{k,j} = Q_{\lambda^{k,j}} + \varepsilon_k \sum_i \lambda_i^{k,j} R_i \longrightarrow M_j = Q_{\lambda^j}. \quad (8.4)$$

The perturbation term tends to zero because its norm is at most $\varepsilon_k \max_i \|R_i\|$. Continuity of ordered eigenvalues shows that M_j has at most one negative eigenvalue. It has at least one, since $f_{\lambda^j}(x^0) < 0$. In particular every limiting multiplier is nonzero and every limiting matrix has exactly one negative eigenvalue.

Choose a unit negative eigenvector $v_{k,j}$ of $M_{k,j}$, oriented so that $v_{k,j}^\top(x^0, 1) > 0$. This scalar product cannot vanish because the quadratic value at $(x^0, 1)$ is negative. Take a further common subsequence so that $v_{k,j} \rightarrow v_j$ for all j . The unique negative eigenvalues converge to the negative eigenvalue of M_j ; passing to the eigenvector equation shows that v_j is its unit eigenvector.

Fix any $x \in S$. For all sufficiently large k , both x and x^0 belong to S_k . The segment joining their lifts is strictly negative for $M_{k,j}$, because the k th aggregation is good on $\text{conv } S_k$. The two scalar products with $v_{k,j}$ therefore have the same sign: otherwise that segment would meet $v_{k,j}^\perp$, where $M_{k,j}$ is positive semidefinite. Hence $v_{k,j}^\top(x, 1) > 0$. Taking limits gives $v_j^\top(x, 1) \geq 0$. Equality is impossible, since $f_{\lambda^j}(x) < 0$ and M_j is positive semidefinite on $v_j^\perp$. Thus every lift of every point of S lies in the same convex negative component of M_j in (8.3). All finite convex combinations of these lifts remain in that component. This proves

$$\text{conv } S \subseteq \bigcap_{j=1}^4 \{x : f_{\lambda^j}(x) < 0\}$$

and proves strict goodness of every limit.

Conversely, a point strictly satisfying all four limiting inequalities strictly satisfies all four perturbed inequalities for sufficiently large k , by (8.4) and finiteness of the index set. It then belongs to $\text{conv } S_k \subseteq \text{conv } S$. This proves (8.1); remove repeated multipliers if desired. No closure is taken in this last argument. $\square$

Remark 8.6 (Dependent triples). The preceding proof already covers dependence. A stronger bound of two follows from the classical two-constraint result and the cone reduction also noted in Blekherman et al. [7, Section 2.5, after Example 2.21]. Indeed, evaluation at a fixed feasible lift is strictly negative on every nonzero nonnegative combination of the Q_i . Their finitely generated cone is therefore pointed. If their span has dimension at most two, intersecting this cone with the affine hyperplane on which that evaluation is -1 gives a compact segment or a point. Its at most two endpoints lie on rays generated by original matrices. Every original inequality is a nonzero nonnegative combination of those at most two inequalities, so they define the same S . Applying Yildiran [26, Theorem 1], with its opposite sign convention, gives at most two good aggregations; translating their multipliers back to the original rows preserves goodness. A single remaining inequality may be duplicated when applying that theorem. This observation is not needed for the four-bound.

8.3 Sharpness and the closed hull

Example 8.7 (The four-necessary example of Blekherman–Dey–Sun). Let $n \geq 3$, $s = x_1$, and $\rho = \sum_{i=2}^n x_i^2$. Consider

$$f_1 = -s^2 + 1 + \rho, \quad f_2 = s^2 + 5s - 4 + \rho, \quad f_3 = -s - \rho. \quad (8.5)$$

This is Blekherman et al. [7, Example 2.21]. The identity $-7f_1 - 3f_2 - 15f_3 = 4s^2 + 5\rho + 5$ proves PDL. The point $(s, \rho) = (-3, 4)$ is strictly feasible. Every feasible point has $s < -1$: $f_1 < 0$ forces $|s| > 1$, while $f_2 < 0$ excludes $s > 1$. Consequently the hull is proper, and the f_1 inequality is good because these points lie in its negative s component. The convex ball inequality $f_2 < 0$ is good as well. The two sums are

$$f_1 + f_3 = -s^2 - s + 1, \quad f_2 + f_3 = s^2 + 4s - 4.$$

The first has one negative homogeneous eigenvalue and is negative on the interval $s < \alpha = (-1 - \sqrt{5})/2$ containing every feasible s . The second is convex and also has one negative homogeneous eigenvalue. Thus the following four rays are good:

$$r_1 = (1, 0, 0), \quad r_2 = (0, 1, 0), \quad r_3 = (1, 0, 1), \quad r_4 = (0, 1, 1).$$

We verify directly that each ray is indispensable. If (a, b, c) is good, then $c \leq a + b$: otherwise the coefficient $a + b - c$ of ρ gives at least $n - 1 \geq 2$ negative eigenvalues. Hence every good multiplier belongs to

$$\{(a, b, c) \geq 0 : c \leq a + b\} = \text{cone}\{r_1, r_2, r_3, r_4\}. \quad (8.6)$$

For the equality, choose $t \in [\max(0, c - b), \min(a, c)]$ and write $(a, b, c) = (a - t)r_1 + (b - c + t)r_2 + tr_3 + (c - t)r_4$. Put $\beta = -2 - 2\sqrt{2}$. The four witnesses below have the displayed slacks in the order $(f_{r_1}, f_{r_2}, f_{r_3}, f_{r_4})$:

designated ray	(s, ρ)	slacks
r_1	$(-2, 3)$	$(0, -7, -1, -8)$
r_2	$(-4, 8)$	$(-7, 0, -11, -4)$
r_3	$(\alpha, 0)$	$(\alpha, 4\alpha - 3, 0, 3\alpha - 3)$
r_4	$(\beta, 0)$	$(4\beta - 3, \beta, 3\beta - 3, 0)$.

Every off-designated slack is strictly negative, since $\alpha, \beta < 0$. Each witness is realizable by setting $x_2 = \sqrt{\rho}$ and the other coordinates to zero. At the witness for r_j , any nonzero multiplier in (8.6) is strictly negative unless it is a positive multiple of r_j . But that witness is outside $\text{conv } S$, since its designated good inequality is zero. Every exact strict-good description must therefore contain all four rays. The upper bound already proved then implies that these four inequalities describe the hull exactly. This proves the sharpness assertion in Theorem 8.1. The argument makes no sharpness claim for $n = 1, 2$.

The strict result also provides a closed conic description, provided the negative components retain their orientation. The conic representation principle is already recorded in Blekherman et al. [7, Remark 2.19]; the bound of four follows from Theorem 8.1.

Corollary 8.8 (Four oriented second-order cone constraints). *Under the hypotheses of Theorem 8.1, choose its multipliers and write $Q_{\lambda^j} = P_j - v_j v_j^\top$ with $P_j \succeq 0$, $P_j v_j = 0$, and $v_j^\top(x^0, 1) > 0$ for one $x^0 \in S$. Then*

$$\text{conv } S = \{x : \|P_j^{1/2}(x, 1)\| < v_j^\top(x, 1) \text{ for } 1 \leq j \leq \ell\}, \quad (8.7)$$

$$\text{cl}(\text{conv } S) = \{x : \|P_j^{1/2}(x, 1)\| \leq v_j^\top(x, 1) \text{ for } 1 \leq j \leq \ell\}. \quad (8.8)$$

Proof. Use the spectral decomposition in (8.3), absorbing the square root of the negative eigenvalue's magnitude into v_j . Strict goodness and connectedness place $\text{conv } S$ in the component containing $(x^0, 1)$. The intersection of those components contains the hull and is contained in (8.1), proving

(8.7). Its closed version contains the closure. Conversely, let x satisfy every closed conic inequality and put $x_t = (1 - t)x + tx^0$, $0 < t \leq 1$. Convexity of the norm gives

$$v_j^\top(x_t, 1) - \|P_j^{1/2}(x_t, 1)\| \geq t(v_j^\top(x^0, 1) - \|P_j^{1/2}(x^0, 1)\|) > 0.$$

Thus $x_t \in \text{conv } S$ and $x_t \rightarrow x$ as $t \downarrow 0$. □

Replacing $<$ by $\leq$ in (8.1) without keeping the orientations need not give this closure. Nor must this closure equal the hull of the original nonstrict set. For example, for $n \geq 2$, take the half-ball example of Blekherman et al. [7, Example 2.23] and append one redundant inequality:

$$f_1 = -x_1^2 + x_1, \quad f_2 = \|x\|^2 - 1, \quad f_3 = -\|x\|^2 - 1.$$

Here $-Q_3 = I$, so PDLC holds, and $S = \text{conv } S = \{x : x_1 < 0, \|x\| < 1\}$. The first two strict inequalities are good and describe S . Their nonstrict counterparts describe the closed half-ball together with the isolated point e_1 . The original nonstrict set contains that same extra point. Its hull is therefore larger than $\text{cl}(\text{conv } S)$.

Theorem 8.1 is an existence result for a fixed finite description. Its proof does not supply an efficient multiplier-selection algorithm or coefficient-size bounds. The exact checks accompanying this section verify the finite identities and witnesses in Example 8.7; the perturbation and limiting arguments are proved above rather than inferred from those checks.

9 Formal verification of the certificate theorem

A Lean development verifies Theorem 2.2 and its HHC specialization, Corollary 2.3, for finite families of real symmetric matrices, real vectors, and real constants. Its strict feasible set, ordinary convex hull, and nontrivial certificate agree with those defined here. The main declaration is `proper_hull_iff_certificate` in the namespace `QuadraticAggregation.System`; the HHC specialization is `proper_hull_iff_certificate_of_hhc`. The assumptions are nonemptiness and AHC, or nonemptiness and HHC, respectively. No positive definite combination is assumed. The certificate-to-proper-hull implication is proved separately without either assumption.

The formal definitions express HHC through kernels of nonzero linear functionals, which are precisely the linear hyperplanes in this finite-dimensional setting. AHC is stated using arbitrarily large good levels, as in Definition 2.1; its equivalence with the increasing sequence formulation is also proved. Matrix positive semidefiniteness is linked to nonnegativity of the quadratic form, using the symmetry of the original matrices and their aggregations.

The proof follows the constant-elimination and coefficient-normalization argument in Section 2. In particular, it constructs separation weights from the convex hyperplane images and obtains the nonzero limit in the actual finitely generated coefficient cone. These are proved steps, not additional hypotheses of the headline theorem. The implementation uses the entrywise maximum matrix norm and the product maximum norm for coefficient pairs. These are equivalent to the Frobenius and Euclidean norms in finite dimension; the compactness proof works in either choice.

The verified supporting results include Lemmas 2.4, 2.5, 2.6, and 2.7, and the uniform cone-separation lemma, Lemma 2.10. The last lemma is checked independently, although the main formal proof does not use it. The spectral distance calculation and quantitative alternative in Section 2.3 are not formalized as separate results.

The sources consist of eleven modules in `Formal/QuadraticAggregation/`, accompanied by a declaration coverage map and reproducible checks. With the pinned Lean 4.33.1 toolchain, targeted compilation without warnings, a transitive axiom audit of all 178 declarations owned by those

modules, and kernel replay of each module passed. The checks were rerun during preparation of this manuscript, with source fingerprints unchanged throughout the run. The only permitted axioms were `propext`, `Classical.choice`, and `Quot.sound`. Kernel replay uses the pinned Lean kernel and imported dependencies; it does not constitute verification by an independently implemented proof assistant or a replay of all dependencies from source.

This formal scope concerns existence of a nontrivial globally convex aggregation. It does not establish that such aggregations describe the entire hull. The closed-system and relaxation consequences, SDP characterizations, examples, applications, further aggregation results, and literature-priority claims are outside the formal package. Formal verification of the stated theorem does not certify the rest of the manuscript.

A An elementary application to local infeasibility certificates

Convex aggregation can also turn a local affine proof into a globally valid convex inequality. We give the exact certificate and a small example to delimit this use. These are applications of established aggregation and convexification principles, not an algorithmic novelty or a performance claim. In related conflict analysis, Berthold and Witzig [4] use local affine proofs and aggregate convex nonlinear relaxation rows. Dong [15] explicitly allows aggregation of original quadratic inequalities or quadratic rows reconstructed from a lift before convexification. The construction below selects nonnegative weights for the original quadratic rows while retaining an affine box margin.

A.1 Validity and a linear selection problem

Let $f_i(x) \leq 0$ be globally active quadratic rows as in (1.1). On a nonempty bounded box $B = [l, u]$, suppose the affine functions $\ell_i(x) = a_i^\top x + d_i$ satisfy $\ell_i(x) \leq f_i(x)$ for all $x \in B$. For $\lambda \geq 0$, put $L_\lambda = \sum_i \lambda_i \ell_i$.

Proposition A.1. *If $\min_{x \in B} L_\lambda(x) \geq \delta > 0$, the box B contains no point satisfying all the original rows. If also $A_\lambda \succeq 0$, then, for every $\bar{x} \in \mathbb{R}^n$,*

$$f_\lambda(\bar{x}) + \nabla f_\lambda(\bar{x})^\top (x - \bar{x}) \leq 0 \tag{A.1}$$

is valid for every original feasible point. It separates every expansion point $\bar{x} \in B$. If $\bar{x}$ minimizes f_λ on B , this one tangent excludes the entire box.

Proof. Original feasibility implies $f_\lambda \leq 0$ everywhere on the feasible set, whereas on B the estimator inequality gives $f_\lambda \geq L_\lambda \geq \delta > 0$. For a convex quadratic, subtraction of its tangent at $\bar{x}$ leaves $(x - \bar{x})^\top A_\lambda (x - \bar{x}) \geq 0$. This proves global validity and separation at any $\bar{x} \in B$. A minimizer on the compact box exists, and the convex first-order condition gives $\nabla f_\lambda(\bar{x})^\top (x - \bar{x}) \geq 0$ for all $x \in B$. Its tangent is therefore at least δ throughout the box. $\square$

An arbitrary tangent separates its expansion point but need not exclude all of B . The quadratic aggregate remains globally valid even though the estimators used to select its weights are local. When several estimators refer to the same original row, their weights must be summed in reconstructing that row's coefficient.

A sufficient positive-semidefiniteness condition is symmetric diagonal dominance with nonnegative

diagonal. Let $a(\lambda) = \sum_i \lambda_i a_i$ and $d(\lambda) = \sum_i \lambda_i d_i$. Consider the linear program

$$\begin{aligned}
& \max_{\lambda, t, v, \delta} \quad \delta \\
& \text{subject to} \quad \lambda \geq 0, \quad \sum_i \lambda_i = 1, \\
& \quad t_{jk} \geq (A_\lambda)_{jk}, \quad t_{jk} \geq -(A_\lambda)_{jk} \quad (j < k), \\
& \quad (A_\lambda)_{jj} \geq \sum_{k < j} t_{kj} + \sum_{k > j} t_{jk} \quad (j = 1, \dots, n), \\
& \quad v_j \leq 0, \quad v_j \leq a_j(\lambda) \quad (j = 1, \dots, n), \\
& \quad \delta \leq d(\lambda) + l^\top a(\lambda) + \sum_j (u_j - l_j) v_j.
\end{aligned} \tag{A.2}$$

The last expression is a lower bound on the exact affine minimum

$$\min_{x \in B} L_\lambda(x) = d(\lambda) + l^\top a(\lambda) + \sum_j (u_j - l_j) \min\{0, a_j(\lambda)\}. \tag{A.3}$$

It attains that minimum by choosing $v_j = \min\{0, a_j(\lambda)\}$. Likewise the absolute-value lifts can be chosen as $t_{jk} = |(A_\lambda)_{jk}|$. Thus (A.2) optimizes exactly the affine margin over the normalized diagonally dominant aggregates. Any feasible solution with certified $\delta > 0$ suffices for validity; no optimality certificate is needed. The elementary identity

$$x^\top A x = \sum_j \left(A_{jj} - \sum_{k \neq j} |A_{jk}| \right) x_j^2 + \sum_{j < k} |A_{jk}| (x_j + \text{sign}(A_{jk}) x_k)^2 \tag{A.4}$$

proves positive semidefiniteness whenever the diagonal dominance conditions hold. Zero off-diagonal terms may be omitted. For rational data and rational weights, all coefficients and the margin can be checked exactly. Diagonal dominance is sufficient, not necessary, and depends on the chosen coordinates.

Affine equalities $h_e(x) = 0$ may be included with signed weights μ_e . Add $\sum_e \mu_e h_e$ to both the original aggregate and its affine estimator. To normalize the selection problem, write $\mu_e = \mu_e^+ - \mu_e^-$, with nonnegative parts, and replace the simplex equation by

$$\sum_i \lambda_i + \sum_e (\mu_e^+ + \mu_e^-) = 1.$$

This permits equality-only certificates and bounds every multiplier. Normalizing only the inequality weights is insufficient: with the row $-1 \leq 0$, equality $x = 0$, and $B = [1, 2]$, the aggregate $-1 + \mu x$ has margin $\mu - 1$ for $\mu \geq 0$. Its margin is unbounded if μ is left unnormalized, although each individual certificate is valid. The equality extension is for affine equalities, whose signed affine estimators are exact.

If original rows are imposed only when specified logical conditions are active, their aggregate is valid under the conjunction of the conditions supporting its nonzero weights. This condition must be retained in any resulting cut. Global unconditional validity in Proposition A.1 uses globally active rows.

A.2 A complete rational example

Let $0 \leq x, y \leq 1$ and impose

$$f_1 = x^2 + 2xy - \frac{3}{50} \leq 0, \quad f_2 = 2y^2 - xy - \frac{9}{100} \leq 0. \tag{A.5}$$

The two quadratic matrices have determinants -1 and $-1/4$, so both are indefinite. The relaxation with square variables s_x, s_y and product variable w imposes the full square envelopes and the four McCormick inequalities,

$$x^2 \leq s_x \leq x, \quad y^2 \leq s_y \leq y, \quad \max\{0, x + y - 1\} \leq w \leq \min\{x, y\},$$

along with $s_x + 2w \leq 3/50$ and $2s_y - w \leq 9/100$. It admits $(x, y, s_x, s_y, w) = (1/5, 1/5, 1/25, 1/25, 0)$. This is the complete termwise convex relaxation on the root box; the failure is not due to missing square tangents.

On the smaller box $B = [1/5, 1]^2$, take

$$\ell_1 = \frac{4}{5}x + \frac{2}{5}y - \frac{9}{50}, \quad \ell_2 = -x + \frac{3}{5}y + \frac{3}{100}.$$

Their validity follows from the polynomial identities

$$\begin{aligned} f_1 - \ell_1 &= (x - 1/5)^2 + 2(x - 1/5)(y - 1/5), \\ f_2 - \ell_2 &= 2(y - 1/5)^2 + (1 - y)(x - 1/5), \end{aligned}$$

whose right sides are nonnegative on B . We have

$$2\ell_1 + \ell_2 = \frac{3}{5}x + \frac{7}{5}y - \frac{33}{100} \geq \frac{7}{100} > 0.$$

The original aggregate is

$$G = 2f_1 + f_2 = 2x^2 + 3xy + 2y^2 - \frac{21}{100} = \frac{1}{2}x^2 + \frac{1}{2}y^2 + \frac{3}{2}(x + y)^2 - \frac{21}{100}.$$

Its quadratic matrix is positive definite, with determinant $7/4$. At $\bar{x} = \bar{y} = 1/5$, the value of G is $7/100$ and its gradient is $(7/5, 7/5)$. Its tangent inequality is exactly

$$x + y \leq \frac{7}{20}. \tag{A.6}$$

It excludes the root relaxation witness and the entire smaller box, where $x + y \geq 2/5 > 7/20$.

The origin and $(1/5, 0)$, $(0, 1/5)$ satisfy both original rows, so the system is globally feasible. With binary variables z_x, z_y and the additional inequalities $x \geq z_x/5$, $y \geq z_y/5$, either single activation is feasible but joint activation is infeasible. Also $\min_B \ell_1 = 3/50 > 0$: a one-row local proof may use only the first, indefinite row, whereas the displayed convex aggregate uses both rows. No general support-selection guarantee follows.

For completeness the selection LP has an exact solution here. Write $\lambda = (a, 1 - a)$, $0 \leq a \leq 1$. Its matrix is

$$A_\lambda = \begin{pmatrix} a & (3a - 1)/2 \\ (3a - 1)/2 & 2(1 - a) \end{pmatrix}.$$

The inequalities $a \geq |(3a - 1)/2|$ and $2(1 - a) \geq |(3a - 1)/2|$ reduce to $1/5 \leq a \leq 5/7$. The affine coefficients of L_λ are $(9a/5 - 1, 3/5 - a/5)$, and its constant is $(3 - 21a)/100$. The second coefficient is positive, while the first changes sign at $a = 5/9$. Formula (A.3) gives

$$\min_B L_\lambda = \begin{cases} (31a - 17)/20, & a \leq 5/9, \\ (11a - 5)/100, & a \geq 5/9. \end{cases}$$

Both pieces are strictly increasing and agree at their common endpoint. Thus the exact optimum is $\lambda = (5/7, 2/7)$ with normalized margin $1/35$. The illustrative weights $(2, 1)$ have normalized margin $7/300$; their unnormalized margin $7/100$ is not comparable to $1/35$.

A.3 Limits of this application

If the quadratic rows are purely bilinear, their quadratic matrices have zero diagonal. Any positive semidefinite aggregate then vanishes: each 2×2 principal minor gives $0 \leq -A_{jk}^2$, hence $A_{jk} = 0$ for every j, k . Signed combinations of purely bilinear equalities have the same obstruction. Cancellation can produce an affine aggregate but cannot produce nonzero positive semidefinite curvature.

Finally every positive semidefinite aggregate is implied by the common Shor lift, by (3.4); its tangents are implied as well. The example demonstrates strength beyond the termwise relaxation, not beyond the full Shor relaxation. The existence theorem does not guarantee a positive-margin diagonally dominant solution on a prescribed local support. No solver implementation or empirical performance conclusion is needed for the validity statements above.

The accompanying script `supplement/check_examples.py` uses rational arithmetic to check the displayed polynomial identities, margins, lifted witnesses, and selected matrix calculations. These checks supplement the proofs; finite test instances do not prove the universal geometric statements.

B Many constraints in a three-dimensional matrix span

The number of generators matters when the homogeneous matrices span a three-dimensional space but more than three constraints are present. This appendix gives an elementary directional refinement of the known facet bound and an obstruction to extending the two-aggregation case to arbitrary polyhedral coefficient cones. These are supporting consequences and examples, rather than a new topological aggregation theorem.

Let $n \geq 3$, $L = \text{span}\{Q_1, \dots, Q_m\}$ have dimension three, and suppose that L contains a positive definite matrix. Assume $S \neq \emptyset$ and $\text{conv } S \neq \mathbb{R}^n$. Define the matrix cone

$$\mathcal{C}_L = \text{cone}\{Q_1, \dots, Q_m\} \subset L, \quad \mathcal{P}_L = L \cap \mathbb{S}_+^{n+1}.$$

We distinguish $\mathcal{C}_L$ from the multiplier cone $\mathcal{K}$ in (1.2). For $M \in L$, write $q_M(x) = (x, 1)^\top M(x, 1)$. Evaluation at a fixed feasible lift is negative on every nonzero nonnegative combination of the generators. It follows that $\mathcal{C}_L$ is pointed. A section on which the negative of that evaluation equals one is the convex hull of the normalized generators, hence is a compact polygon. If $\mathcal{C}_L$ has k extreme rays, it consequently has k facets, each generated by two extreme rays. We may retain one original matrix on each extreme ray: every omitted row is a nonzero nonnegative combination of the retained rows, so the strict set and the available aggregation matrices are preserved.

Blekherman, Dey, and Sun give the resulting $2k$ bound [7, Proposition 2.22]. Their proof uses the facets of this three-dimensional cone. The next proposition records how the count depends on the chosen positive definite direction; its three-generator antecedent is Blekherman and Dunbar [5, Proposition 8.6].

Proposition B.1 (Directional facet bound). *Write an irredundant facet representation*

$$\mathcal{C}_L = \{M \in L : \ell_j(M) \geq 0, j = 1, \dots, k\}.$$

For every $D \in L$ with $D \succ 0$, put $J_-(D) = \{j : \ell_j(D) < 0\}$. Then $\text{conv } S$ is described by at most $2|J_-(D)|$ good aggregations. In particular,

$$\mathcal{P}_L \not\subseteq -\mathcal{C}_L \implies \text{at most } 2k - 2 \text{ good aggregations suffice.} \quad (\text{B.1})$$

Proof. First, the full system has HHC. Choose three basis matrices for L . On each homogeneous hyperplane their restrictions still admit a positive definite real combination. That hyperplane has

dimension $n \geq 3$, so the three-form convexity theorem [19, Theorem 2.1] gives a convex joint image. The image of the original m forms is a linear image of this convex image. The full good-aggregation description therefore applies [7, Theorem 2.9].

Since $D \succ 0$ has positive evaluation at a feasible lift, it cannot belong to $\mathcal{C}_L$. Thus $J_-(D)$ is nonempty. Let $M \in \mathcal{C}_L$ be a good aggregation matrix whose strict sublevel is not all of $\mathbb{R}^n$. It has exactly one negative eigenvalue. Set

$$t_* = \min_{j \in J_-(D)} \frac{\ell_j(M)}{-\ell_j(D)}, \quad M_* = M + t_* D.$$

Every facet inequality remains satisfied up to t_* , and at least one indexed by $J_-(D)$ becomes an equality. Hence $M_* \in \mathcal{C}_L$ lies on a selected facet. Also $M_* \neq 0$: for $t_* = 0$ this follows from $M \neq 0$, and for $t_* > 0$ the alternative would make $M = -t_* D$ negative definite, contrary to its inertia.

The positive semidefinite improvement lemma [7, Proposition 9.1] shows that M_* is good and $\{q_{M_*} < 0\} \subseteq \{q_M < 0\}$. To check its coefficient hypothesis, choose nonnegative representations $M = Q_\lambda$ and $M_* = Q_\mu$. Then $Q_{\mu-\lambda} = t_* D \succeq 0$ and $\lambda + (\mu - \lambda) = \mu \succeq 0$, with $\mu \neq 0$. Thus every nonredundant good cut is dominated by a good cut on a selected facet. The intersection of all good cuts on these facets equals the hull: it contains the hull by goodness and is contained in every original good cut by this domination.

On each facet the available matrices are nonnegative combinations of its two extreme generators. The pair-support reduction [7, Proposition 9.6] describes the intersection of its good cuts using at most two such cuts. Goodness here is relative to the full set S , as in that proposition, not to the two-row subsystem. Facets without good cuts contribute no inequality. Summing over the selected facets proves $2|J_-(D)|$.

Finally, if $E \in \mathcal{P}_L \setminus (-\mathcal{C}_L)$, some $\ell_j(E) > 0$. Choose $D_0 \succ 0$ in L . For sufficiently small $\delta > 0$, $D = E + \delta D_0 \succ 0$ and $\ell_j(D) > 0$. Thus $|J_-(D)| \leq k - 1$, proving (B.1). $\square$

Minimizing $|J_-(D)|$ over positive definite $D \in L$ can strengthen the displayed bound for a particular cone. This is an existence bound; no procedure for finding a best direction or the endpoint multipliers is asserted. The proof uses strict aggregation directly and requires no regularity or boundedness assumption on the original weak feasible set.

Example B.2 (An arbitrarily large necessary family of ellipsoids). Let $m \geq 3$, $n \geq 2$, and $0 < a_1 < \dots < a_m < 1$. Put $\rho = \sum_{j=2}^n x_j^2$ and define

$$f_i(x) = a_i x_1^2 + (1 - a_i^2)\rho - 1, \quad U = \{x : f_i(x) < 0 \text{ for all } i\}. \quad (\text{B.2})$$

Then U is nonempty, open, bounded, and convex, since its defining quadratic parts are positive definite and the origin is strictly feasible. In the basis (x_1^2, ρ, t^2) of the homogeneous span, the matrices have coordinate vectors $(a_i, 1 - a_i^2, -1)$. For $a < b < c$,

$$\det \begin{pmatrix} a & 1 - a^2 & -1 \\ b & 1 - b^2 & -1 \\ c & 1 - c^2 & -1 \end{pmatrix} = (b - a)(c - a)(c - b) > 0. \quad (\text{B.3})$$

This follows by the elementary column operations that turn the matrix into a Vandermonde matrix. Hence the span is exactly three-dimensional and contains the positive definite form $x_1^2 + \rho + t^2$.

For each i , choose p_i with

$$(p_i)_1^2 = \frac{2a_i}{1 + a_i^2}, \quad \sum_{j=2}^n (p_i)_j^2 = \frac{1}{1 + a_i^2}. \quad (\text{B.4})$$

For example, take the positive square roots in coordinates one and two and zeros elsewhere. Direct substitution gives the identity

$$f_j(p_i) = -\frac{(a_j - a_i)^2}{1 + a_i^2}. \quad (\text{B.5})$$

Thus, at p_i , a nonzero nonnegative aggregation is nonnegative exactly when its multiplier is a positive multiple of e_i . Since $p_i \notin U = \text{conv } U$, every exact strict aggregation family, even an infinite one, must contain that ray. The original m rows are good, so the minimum finite cardinality is exactly m . Each row has exactly one negative homogeneous eigenvalue. The witness also exposes its matrix ray in $\mathcal{C}_L$: all other generators have strictly negative evaluation at p_i . In particular $k = m$.

The same necessity holds for finite nonstrict aggregation descriptions of $\text{cl } U$. Indeed, if a finite family omits ray i , all its inequalities are strict at p_i and remain satisfied in a neighborhood. For $t > 1$ close to one, $f_i(tp_i) = t^2 - 1 > 0$, so that neighborhood contains a point outside $\text{cl } U$. Conversely the original m weak inequalities describe $\text{cl } U$: shrinking any weak feasible point toward the origin makes every inequality strict. Thus m is also the minimum finite weak count. No indispensability assertion is made for infinite weak families, where a missing ray can be approached by limits.

The complementary geometry $\mathcal{P}_L \subseteq -\mathcal{C}_L$ does not imply a bound independent of k . In particular the two-aggregation conclusion of Blekherman and Dunbar [5, Proposition 8.10], proved for a three-generator cone under its closed-set hypotheses, fails for general polyhedral cones even when those geometric hypotheses hold.

Proposition B.3 (Obstruction in the complementary cone case). *For every $m \geq 3$ and $n \geq 3$, there is a strictly feasible system of $m + 2$ quadratic inequalities whose three-dimensional homogeneous span contains a positive definite matrix, for which $\mathcal{P}_L \subseteq -\mathcal{C}_L$ and $k = m + 2$, but every exact strict aggregation description of its hull needs the $m = k - 2$ specified rays. Exactly m good aggregations suffice. Its original weak feasible set is compact, has nonempty interior, equals the closure of its interior, and has no points at infinity. Its minimum finite weak aggregation count is also exactly m .*

Proof. Start with (B.2) and append the two inequalities

$$g_1(x) = -x_1^2 < 0, \quad g_2(x) = -\rho < 0.$$

The new strict set is

$$V = U \cap \{x : x_1 \neq 0, (x_2, \dots, x_n) \neq 0\}.$$

The added rows change the strict set, but $\text{conv } V = U$. To see this, fix $x \in U$ and choose a ball about x contained in U . A sufficiently small vector z can be chosen with $z_1 \notin \{x_1, -x_1\}$ and $(z_2, \dots, z_n) \notin \{(x_2, \dots, x_n), -(x_2, \dots, x_n)\}$. Both $x + z$ and $x - z$ then belong to V . Thus x is their midpoint. The reverse inclusion follows from convexity of U . This also proves nonemptiness and properness of the new hull.

In the coordinate basis (x_1^2, ρ, t^2) , let E_1, E_2, E_3 be the three basis matrices. The new matrix cone is

$$\mathcal{C}_L = \text{cone}\{Q_1, \dots, Q_m, -E_1, -E_2\}.$$

At every p_i from (B.4), both added rows are strictly negative, while (B.5) is unchanged. Each Q_i therefore still spans an exposed extreme ray. The rays $-E_1, -E_2$ are also extreme: every Q_i has t^2 coefficient -1 , so a nonnegative representation of a matrix with zero t^2 coefficient can use only $-E_1, -E_2$. These two rays are independent and extreme in their two-dimensional cone. Since all generators have now been shown extreme and distinct, $k = m + 2$.

For every i the exact identity

$$Q_i + a_i(-E_1) + (1 - a_i^2)(-E_2) = -E_3 \quad (\text{B.6})$$

places $-E_3$ in $\mathcal{C}_L$. A matrix $uE_1 + vE_2 + wE_3$ is positive semidefinite exactly when $u, v, w \geq 0$. Hence $-\mathcal{P}_L$ is the cone of $-E_1, -E_2, -E_3$, and (B.6) proves $\mathcal{P}_L \subseteq -\mathcal{C}_L$.

Any nonzero nonnegative aggregation of the enlarged system that is not supported only on original row i is strictly negative at p_i . That point is excluded from $\text{conv } V = U$. Therefore every exact strict aggregation description must contain each of the m original rays, even if additional infinitely many aggregations are allowed. Conversely, the original m convex inequalities are good for V and describe its hull. They attain the bound m .

The added weak inequalities hold globally. The original weak set is therefore still $C = \{f_i \leq 0 \ \forall i\} = \text{cl } U$. It is compact, has interior U , and equals the closure of that interior. For the interior equality, a point with $f_i = 0$ lies on the boundary of its positive definite ellipsoid and cannot be interior to C . Each original leading matrix is positive definite, so a common nonpositive direction at infinity must be zero. The same finite-family neighborhood argument at p_i proves that a weak aggregation description of C needs all m original rays. They give the matching description. $\square$

Thus the complementary hypothesis alone does not support a general-cone two-bound, even for regular compact weak systems. This disproves that possible shortcut to improving the facet count. It does not disprove an unconditional $2k - 2$ bound: the necessary count $k - 2$ in this example is smaller. Our established general upper bound remains the cited $2k$, with (B.1) and the directional count when their hypotheses apply. No optimal count for arbitrary cones is asserted. Nor can one obtain such a count simply by triangulating the coefficient cone and intersecting subsystem hulls: convex hull does not commute with intersection, and goodness is relative to the original full feasible set. The inertia condition also fails to be closed under addition: the two matrices $\text{diag}(-4, 1, 1, 0)$ and $\text{diag}(1, -4, 1, 0)$ each have one negative eigenvalue, whereas their sum has two. Consequently their individual inertia bounds cannot justify a reduction that adds good matrices arbitrarily.

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